

How Much Does Air Connectivity Increase Aviation CO₂?

[author order TBD]

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Abstract

Air connectivity is an explicit object of national policy, yet its carbon consequences have never been estimated causally at the global level. Here we construct flight-stage-resolved CO₂ for more than 6,000 airports over 1996–2023 from schedule data, match it to a global panel of airport connectivity, and identify the effect of connectivity on emissions with an instrument based on each country’s air-versus-sea market-access geography. A one percent increase in connectivity raises national aviation CO₂ by 5.3 percent, an elasticity far above one and invariant to how emissions are allocated across countries. Decomposing the response on the accounting identity shows that it is a frequency response: flights rise by 6.8 percent, aircraft size and stage length do not rise, and the decline in emissions per seat-kilometre is small (0.3 percent) and imprecisely estimated, offsetting at most a small fraction of the scale effect. The elasticity is concentrated in low-income, newly connecting countries and describes the take-off stage of network development, in level as well as in logarithmic specifications; at the airport level it is systemic rather than hub-driven, although the emissions it generates are more concentrated than emissions themselves and are led by new hubs. Emissions also respond to the connectivity of contiguous neighbours: across spatial lag, error and Durbin specifications the neighbour elasticity is 1.1 to 2.1, the spatial lag of emissions itself is negligible, and the total effect of a network-wide connectivity gain is 5.6, the same as the single-equation elasticity. Connectivity growth since 1996 accounts for 41.1 percent (349 Mt) of 2023 aviation CO₂, an annual external cost of \$18 billion to

\$66 billion at standard social-cost-of-carbon values, and only sustained fuel switching, not the efficiency gains of network development, materially lowers this level.

Main

Air connectivity has become an important part of economic and transport policy. Governments expand airport capacity, support new routes and negotiate air-service agreements to improve access to the global air network (Fageda et al., 2018; Piermartini and Rousová, 2013). A large body of literature links better connectivity to trade, tourism, investment, and regional development (Lenaerts et al., 2021; Zhang and Graham, 2020), making network expansion particularly important for countries with limited global access. However, the climate implications of air connectivity development are less clear. Aviation remains difficult to decarbonize because large-scale electrification is not expected in the near term, while continued growth in air travel can offset gains from more efficient aircraft and lower-carbon fuels (Bergero et al., 2023; Sacchi et al., 2023). As countries continue to expand their air networks, a basic question emerges: what happens to carbon emissions when air connectivity improves?

The answer depends on how connectivity changes aviation activity. On the one hand, better connections can open new destinations and induce trips that would otherwise not occur (Koo et al., 2017; Piermartini and Rousová, 2013). On the other hand, they can also change how those trips are served. Changes in network structure can alter flight frequency, aircraft size, and routing, with potentially offsetting effects on operating and environmental efficiency (Brueckner, 2004; Pels, 2021). These changes can work in opposite directions. More aviation activity raises total emissions, while greater efficiency lowers the emissions associated with each unit of travel. Evidence from other transport settings shows that induced demand can offset some of the environmental gains from improved efficiency (Hymel et al., 2010; Ou et al., 2024). The net carbon effect of connectivity is therefore an empirical question. So is the source of that effect: whether emissions change because airlines operate more flights, use different aircraft, fly different distances, or provide transport more efficiently.

Existing research has not resolved this question. Studies of air connectivity have focused mainly on its economic benefits, linking better access to trade, tourism, productivity and regional development (Lenaerts et al., 2021; Zhang and Graham, 2020), while research on aviation decarbonization has focused on travel demand, aircraft and operational efficiency, and lower-carbon fuels (Bergero et al., 2023; Dray et al., 2022). Much less is known about how the development of the air network itself shapes aviation activity and emissions. Estimating this effect is difficult because causality can run in both directions. For example, while airlines add routes and flights in response to growing demand, better connections can also generate additional travel (Koo et al., 2017; Pitfield et al., 2010). Income, trade, and tourism can further affect both network expansion and aviation activity (Van De Vijver et al., 2014). These sources of endogeneity can bias estimates based on observed changes in connectivity. Therefore, identifying the average causal effect of connectivity requires variation that is not driven by local travel demand. This gap in causal evidence on the carbon effects of air connectivity motivates our study.

However, the average effect alone provides an incomplete picture of the carbon consequences of air connectivity. First, the carbon effect may change as air networks develop. New connections can generate substantial new activity in countries with sparse networks, while the same increase in connectivity may have a smaller effect once networks are well developed. Therefore, the carbon effect of connectivity may depend on the stage of network development. Second, the effects may extend beyond the country where connectivity expands. Air transport operates as an international network, so greater connectivity in one country can change aviation activity in neighboring countries. Emissions associated with network growth may also be concentrated in particular airports rather than spread evenly across the network. These differences matter for understanding when and where the carbon consequences of network expansion are greatest.

In this study, we address three questions. First, how does air connectivity affect aviation CO₂, and through which changes in aviation activity and efficiency? Second, how does the effect change as air networks develop? Third, how are the carbon effects of connectivity distributed across the air network, and do they extend across national borders? We answer these questions using a global panel of air connectivity and flight-stage-resolved CO₂ emissions from more than

6,000 airports between 1996 and 2023. To address endogeneity, we use a shift-share instrumental-variable design based on countries’ predetermined exposure to global aviation expansion.

Our results reveal a large positive effect of air connectivity on aviation CO₂, particularly during the “take-off” stages of network development. Our main estimate indicates that a 1% increase in connectivity increases aviation CO₂ by about 5.3%, driven primarily by a roughly 7% increase in flight frequency. Aircraft size, flight distance, and carbon intensity decline slightly, but these changes are not statistically significant. The effect on emissions is largest in countries with lower initial income and connectivity and declines as networks mature. The effects also extend across national borders, while emissions associated with network growth are concentrated in a small share of airports. These findings show that network expansion increases aviation emissions mainly by generating additional flying, while accompanying efficiency gains are too small to offset this growth. Therefore, accounting for network-driven growth in aviation activity is important for assessing pathways to aviation decarbonization.

Results

The emissions elasticity of connectivity is far above one

A one percent increase in air connectivity raises national aviation CO₂ by 5.35 percent (Table 1). The elasticity is far above one, the benchmark at which emissions grow in proportion to the network, and the difference from one is statistically significant. Three features of the estimate matter for what follows. First, it is not an artefact of measurement choices. The elasticity remains above one under every aggregation of airport connectivity to the country level (Table 1, Panels C–E) and moves little across emission allocation rules, from 5.30 under a 50/50 split of cruise emissions to 5.70 under the territorial rule (Extended Data Table 2). Ordinary least squares estimates are about one third smaller, consistent with attenuation from measurement error in the connectivity index. Second, it is not a development effect in disguise. Adding log GDP, trade openness, urbanisation, foreign direct investment and tourist arrivals, all themselves

outcomes of connectivity, moves the elasticity only to between 4.4 and 4.6 (Extended Data Table 11). Third, it is not constant over time. The elasticity is 5.61 for 1996–2007 and 2.95 for 2010–2023 excluding the COVID-19 years, so it halves between the network-expansion and mature-network eras but remains above one in both (Extended Data Table 1 and Extended Data Fig. 1). Inference is described in Methods.

The same variation raises seat-kilometres by 5.64 percent, slightly faster than emissions, so emissions per seat-kilometre fall by 0.29 percent, an efficiency term not distinguishable from zero (Table 2). Emissions therefore grow with the network almost exactly as traffic does.

Table 1: Air connectivity and aviation CO₂: main results

	Bunker CO ₂ (1)	Intl. CO ₂ (2)	Seat-km (3)
<i>Panel A. OLS</i>			
ln GACI (cwm)	3.388*** (0.364)	3.526*** (0.358)	3.684*** (0.398)
<i>N</i>	4,634	4,634	4,634
<i>Panel B. 2SLS, Feyrer instrument, capacity-weighted mean</i>			
ln GACI (cwm)	5.347*** (1.076)	5.192*** (1.043)	5.641*** (1.194)
KP <i>F</i>	18.4	18.4	18.4
<i>N</i>	4,634	4,634	4,634
<i>Panel C. 2SLS, GACI sum</i>			
ln GACI (sum)	2.153*** (0.524)	2.091*** (0.542)	2.272*** (0.592)
KP <i>F</i>	14.0	14.0	14.0
<i>N</i>	4,634	4,634	4,634
<i>Panel D. 2SLS, GACI max</i>			
ln GACI (max)	4.548*** (0.841)	4.416*** (0.845)	4.798*** (0.955)
KP <i>F</i>	21.0	21.0	21.0
<i>N</i>	4,634	4,634	4,634
<i>Panel E. 2SLS, GACI mean</i>			
ln GACI (mean)	5.562*** (1.389)	5.401*** (1.334)	5.868*** (1.491)
KP <i>F</i>	29.3	29.3	29.3
<i>N</i>	4,634	4,634	4,634

Notes: Country-year panel, 1996–2023. All specifications include country and year fixed effects, log population and log sea market access. Outcomes are in logs: total aviation CO₂ under the bunker convention (all cruise plus departure-side LTO assigned to the departure country), bunker CO₂ from international flights, and scheduled departing seat-kilometres; CO₂ per seat-km is in Table 2. Panels B–E instrument log GACI with the Feyrer interaction of world aviation technology and 1996 air-versus-sea geography (Methods). Standard errors clustered by country in parentheses; the Kleibergen–Paap *F* is cluster-robust. ***, **, *: significance at 1, 5, and 10 percent.

Where the elasticity comes from: frequency, not aircraft size or efficiency

The response is a frequency response (Table 2 and Extended Data Fig. 2). Because national CO₂ is identically the product of flights, seats per flight, kilometres per flight and CO₂ per seat-kilometre (Methods, equations 23 and 24), the elasticities of the four components sum to the

total and show where the 5.35 percent arises. Flights rise by 6.76 percent for a one percent gain in connectivity, whereas aircraft gauge (-0.34), stage length (-0.78) and emissions per seat-kilometre (-0.29) do not change significantly. Scale, the sum of the first three terms, accounts for 106 percent of the total and the efficiency term offsets 6 percent. Composition margins are negligible: neither the landing-and-take-off share of emissions nor the international share of seat-kilometres moves. Causal mediation on the same data attributes 84 to 85 percent of the total effect to flight frequency and seat-kilometres and none to the international share (Extended Data Table 4).

Table 2: Decomposition of the CO₂ elasticity on the accounting identity

	Elasticity to GACI	s.e.	Share of total (%)
<i>Scale components</i>			
Flights (frequency)	+6.760***	(1.367)	+126
Seats per flight (aircraft gauge)	-0.336	(0.432)	-6
Km per flight (average stage length)	-0.782	(0.569)	-15
Subtotal: scale (seat-km)	+5.641***	(1.194)	+106
<i>Efficiency component</i>			
CO ₂ per seat-km (carbon intensity)	-0.294	(0.286)	-6
Total CO₂ (equals Table 1, column 1)	+5.347***	(1.076)	+100
<i>Composition margins (memo)</i>			
LTO share of CO ₂	+0.037	(0.069)	–
International share of seat-km	+0.019	(0.115)	–

Each row is a separate 2SLS regression of the log component on log GACI (capacity-weighted mean), instrumented by the Feyrer interaction, with log population and log sea market access, country and year fixed effects, and standard errors clustered by country; identical sample of 4,634 country-years, first-stage Kleibergen–Paap $F = 18.4$. Because CO₂ = flights $\times$ seats per flight $\times$ km per flight $\times$ CO₂ per seat-km, the log elasticities add exactly to the total. Load factor is constant by construction (scheduled capacity) and cancels. Shares are component elasticity divided by the total. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The decomposition varies by income, although the within-tercile estimates are imprecise (Extended Data Table 5). In the bottom income tercile every scale margin expands: flights by 8.5 percent, aircraft size by 0.6 and stage length by 1.1, with intensity falling by 0.7 percent, for a total of 9.4. In the middle tercile flights rise by 8.6 percent but stages shorten, for a total of 6.1. In the top tercile the total is 1.9 and no component is individually significant, although the point estimates indicate that frequency still rises while aircraft become smaller and stages shorter. Applied separately to international and domestic departures, the identity shows

that the international elasticity (5.19) is carried by frequency (5.59), with slight up-gauging (0.52) and an intensity decline (-0.46), while domestic emissions do not respond. A secondary instrument whose compliers are hub-quality changes rather than market-access growth gives the complementary picture: no response of total emissions, but a 3.62 percent rise in international emissions and a 1.21 percent rise in emissions per seat-kilometre, the latter significant at the 10 percent level (Methods). Additional flights, not larger aircraft or longer routes, therefore carry the response.

The elasticity is concentrated in newly connecting countries

The elasticity falls steeply with 1996 income and connectivity (Fig. 1 and Extended Data Table 3). By income, the elasticity of total CO₂ is 9.43 in the bottom tercile, 6.12 in the middle and 1.89 at the top, and the pooled interaction of the top tercile with connectivity is -3.26 . By baseline connectivity it falls from 7.48 in the bottom tercile to 5.07 in the middle; the top tercile has no first stage, a fact used as a diagnostic of the exclusion restriction (Methods). By region, the elasticity is 5.05 in Asia-Pacific, 4.54 in Africa and 2.93 in Latin America, and indistinguishable from zero in Europe (-0.47). The efficiency margin follows the same geography, with the largest decline in emissions per seat-kilometre in the bottom income tercile (-0.73), but no tercile estimate is significant. Because the split-sample estimates are less precisely identified than the pooled one, the pooled interaction is the more reliable statement of the gradient. The headline elasticity is therefore driven by countries with low baseline income and connectivity, and it is smaller in mature networks for the reason the decomposition makes explicit: in high-income networks the point estimates indicate that additional frequency is offset by smaller aircraft and shorter stages.

What the elasticity represents: the take-off stage of network development

The headline number describes the take-off stage of network development, for three reasons. First, the gradient is not an artefact of the logarithmic scale. Re-estimated with connectivity in levels (Extended Data Table 12), the semi-log slope is 3.86 log points of CO₂ per unit of GACI, an elasticity of 4.2 at the sample mean, and the level slope is 4.3 Mt per unit; within baseline-connectivity terciles the semi-log slope is 9.1, 4.6 and 1.1 from bottom to top, so the same absolute addition to the network raises emissions far more where the network is thin. Second, the gradient is a step rather than a smooth slope, and it is not confined to the very bottom of the distribution (Extended Data Fig. 3 and Extended Data Table 13). Quintile splits give elasticities of 7.4 and 8.5 in the two lowest quintiles, 3.3 in the third and 4.0 in the fifth, with the fourth unidentified. A pooled specification that interacts connectivity with centred log baseline connectivity gives a fitted elasticity of 6.5 at the tenth percentile of baseline connectivity, 5.6 at the median and 4.1 at the ninetieth, and the quadratic term indicates a floor rather than a continued decline. Third, the identifying variation is concentrated in the expansion era, and the elasticity falls from 5.6 to 2.9 between the two eras. The headline elasticity therefore describes the emissions response during the take-off stage of national network development, which the sample observes for most countries between 1996 and the late 2000s, rather than a universal constant.

This reading matters for extrapolation. A single elasticity applied to every country, as in the attribution below, overstates the contribution of mature networks and understates that of new entrants. With tercile-specific elasticities, the fitted elasticity at each country's baseline, or the semi-log slope applied to the absolute change in connectivity, the world total is 294 to 346 Mt, or 35 to 41 percent of 2023 emissions, against 349 Mt under the common elasticity (Extended Data Table 14). The headline attribution is thus the upper end of a range whose lower end is about one third of current emissions.

Heterogeneity of the connectivity elasticity (filled = $p < 0.05$, bars = 95% CI)

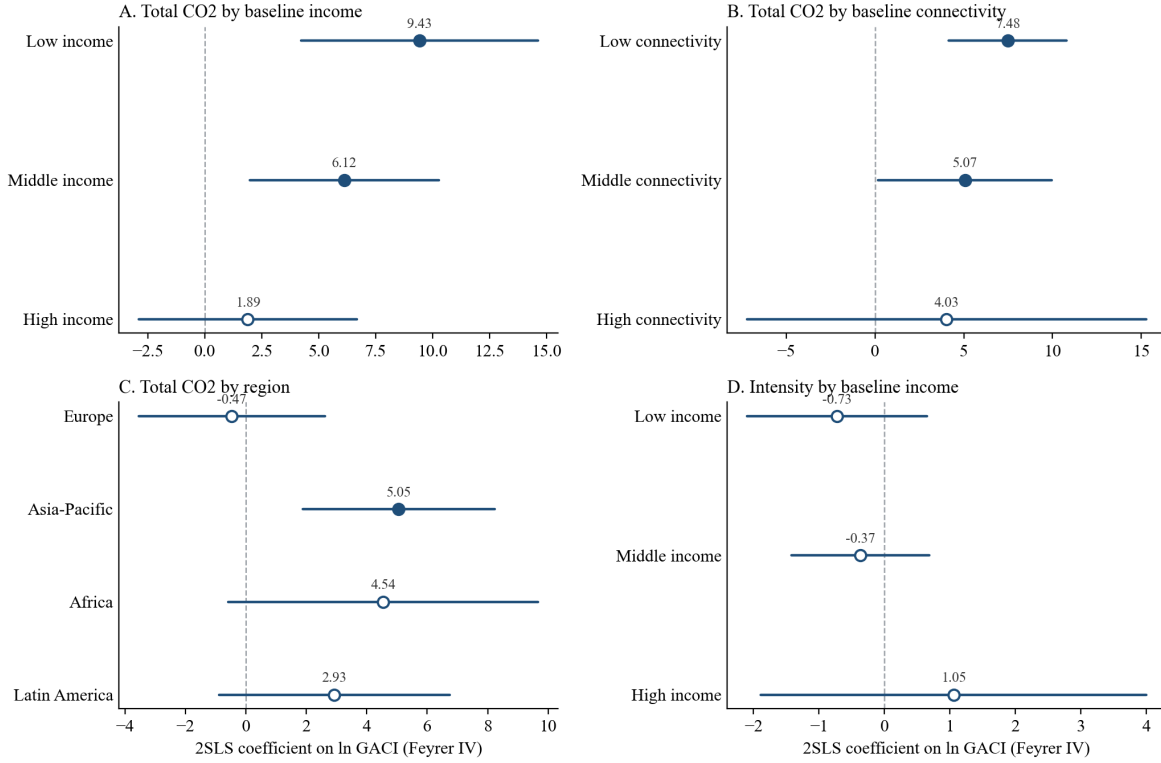


Figure 1: **Heterogeneity of the connectivity elasticity** (2SLS, Feyrer instrument; total bunker CO₂ in panels A–C, CO₂ per seat-km in panel D). Filled markers denote $p < 0.05$; bars are 95 percent confidence intervals. The Middle East and North America are omitted as unidentified.

Emissions respond to neighbours' connectivity, not to neighbours' emissions

Emissions also respond to the connectivity of contiguous neighbours (Table 3). The table nests the single-equation estimate in five spatial specifications: the spatial lag of connectivity (SLX), the spatial lag of the outcome (SAR), a spatial error (SEM) and the Durbin combinations (SDM and SDEM), with own and neighbours' connectivity and the spatial lag of emissions instrumented in Panel B (Methods). Contiguity is the only neighbour definition under which the spillover survives a permutation placebo that relabels countries at random in the weight matrix (Extended Data Tables 8 and 9; Extended Data Figs. 10 and 11; inverse-distance weights in Extended Data Table 10).

Three results follow. First, the spillover runs through neighbours' connectivity rather than through neighbours' emissions or common shocks. The spatial lag of CO₂ is 0.03 in the SAR and -0.02 in the SDM, and the spatial-error parameter is 0.09 to 0.11, whereas the neighbours' connectivity term is 1.85 in the SLX, 1.09 in the SDM and 2.08 in the SDEM. Second, the own elasticity is robust to the spatial specification: 5.43 under the spatial lag, 4.80 under the spatial error and 4.78 as the direct effect of the SDM, against 5.35 in Table 1, and 2.95 to 3.22 once the neighbours' term absorbs the part of the own response that moves with neighbours' growth. Third, the total effect of a network-wide increase in connectivity, direct plus indirect, is 5.57 in the SDM, the same as the single-equation elasticity; its indirect component, 0.79, is the cross-border part. The margins of the spillover differ from those of the own response: a neighbour's connectivity gain raises own seat-kilometres by 2.45 percent through more international flying and lowers emissions per seat-kilometre by 0.60 percent, the signature of traffic feeding a regional hub, whereas the own response works through frequency (Extended Data Table 8, Panel B). National estimates therefore understate the network-wide emissions effect of connectivity growth by 15 to 57 percent of the own effect, depending on whether the SDM indirect effect or the SLX neighbour elasticity is used, and the emissions accrue across borders.

Table 3: Spatial models of the connectivity elasticity: SLX, SAR, SEM and SDM

	OLS-FE (1)	SLX (2)	SAR (3)	SEM (4)	SDM (5)	SDEM (6)
<i>Panel A. Connectivity treated as exogenous (OLS; ML for the spatial models)</i>						
ln GACI (own)	3.39*** (0.36)	3.29*** (0.38)	3.33*** (0.06)	3.35*** (0.06)	3.28*** (0.06)	3.31*** (0.06)
W ln GACI (neighbours)	—	0.65** (0.27)	—	—	0.48*** (0.10)	0.66*** (0.09)
W ln CO ₂ (spatial lag, ρ)	—	—	0.12*** (0.02)	—	0.07*** (0.02)	—
W u (spatial error, λ)	—	—	—	0.12*** (0.02)	—	0.12*** (0.02)
Direct effect	—	—	3.34*** (0.06)	—	3.29*** (0.06)	—
Indirect effect	—	—	0.34*** (0.05)	—	0.61*** (0.08)	—
Total effect	—	—	3.68*** (0.08)	—	3.91*** (0.09)	—
	2SLS	SLX-IV	SAR-IV	SEM-IV	SDM-IV	SDEM-IV
<i>Panel B. Own connectivity instrumented (generalised spatial 2SLS)</i>						
ln GACI (own)	5.35*** (1.07)	3.22*** (1.23)	5.43*** (0.93)	4.80*** (1.13)	4.78*** (0.97)	2.95*** (0.99)
W ln GACI (neighbours)	—	1.85*** (0.66)	—	—	1.09* (0.64)	2.08*** (0.64)
W ln CO ₂ (spatial lag, ρ)	—	—	0.03 (0.03)	—	-0.02 (0.01)	—
W u (spatial error, λ)	—	—	—	0.09	—	0.11
Direct effect	—	—	5.43*** (1.01)	—	4.78*** (0.92)	—
Indirect effect	—	—	0.13 (0.19)	—	0.79* (0.43)	—
Total effect	—	—	5.56*** (1.07)	—	5.57*** (0.91)	—
First-stage F , own connectivity	18.5	7.2	4.7	7.7	6.2	5.2
N	4,634	4,634	4,634	4,634	4,634	4,634

Notes: Outcome is log bunker CO₂; specification as in Table 1, column 1. W is the row-normalised land-contiguity matrix applied within each year; the 38 countries without a land neighbour have a zero row. SLX adds neighbours' log GACI, SAR the spatial lag of the outcome, SEM a spatially autoregressive error, SDM and SDEM the respective combinations. Panel A is estimated by maximum likelihood on the within-transformed data with conventional (unclustered) standard errors; Panel B by generalised spatial 2SLS with the Feysr shifter, its spatial lags and the spatial lags of the controls as instruments, standard errors clustered by country (Methods). Direct, indirect and total effects and their standard errors are computed as described in Methods. First-stage F is the Sanderson–Windmeijer statistic for own connectivity. ***, **, *: significance at 1, 5, and 10 percent.

Connectivity growth accounts for 41.1 percent of 2023 emissions

Applying the estimated elasticity to each country's change in connectivity since 1996 attributes 349 Mt, or 41.1 percent, of 2023 aviation CO₂ to post-1996 network expansion (Fig. 2; Methods).

The attributed growth is concentrated in emerging hub countries: China (+85.9 Mt), the United States (+34.5), the United Arab Emirates (+24.5), Japan (+16.9), Turkey (+16.5), India (+15.4) and Korea (+14.1), while Germany’s declining connectivity contributes -14.7 Mt (Table 4 and Extended Data Fig. 4). With elasticities that vary by baseline connectivity the world total is 294 to 346 Mt (Extended Data Table 14). At standard social-cost-of-carbon values these emissions represent an annual external cost of \$17.8 billion to \$66.3 billion, depending on whether the US Interagency Working Group interim value (\$51 per tonne), the estimate of Rennert et al. (\$185) or the US EPA central value (\$190) is applied (Interagency Working Group on Social Cost of Greenhouse Gases, 2021; Rennert et al., 2022; US Environmental Protection Agency, 2023). At the EPA value the cost is \$16.3 billion per year for China, \$6.6 billion for the United States and \$4.6 billion for the United Arab Emirates, and Germany’s decline corresponds to an avoided cost of \$2.8 billion per year.

Where these emissions are generated is not where they are booked. The bunker-attributed share of world emissions exceeds the physical landing-and-take-off share by up to 1.4 percentage points where stage lengths are long, at the Gulf hubs and in the United States, and falls short by up to 3.5 points where networks are dominated by short-haul domestic flying, most of all in China (Extended Data Figs. 5 and 6). Any scheme that allocates responsibility by booked emissions, including CORSIA, embeds a transfer across these accounting boundaries.

The country estimates are not an artefact of national aggregation. In an airport panel with airport and country-by-year fixed effects, a one percent increase in an airport’s connectivity is associated with 3.76 percent more CO₂, a 0.50 percent decline in CO₂ per seat-kilometre and a 0.34 point higher international share. The elasticity is 2.0 to 2.1 at airports in the 1996 global top five percent and 3.8 to 4.1 elsewhere, so the response is systemic rather than hub-driven (Extended Data Table 6 and Extended Data Fig. 7; these estimates are descriptive, as no valid airport-level instrument is available, Methods). The emissions the response generates are nonetheless concentrated. Applying the elasticity to each airport’s connectivity change attributes 346 Mt across airports, of which the top five percent of airports, some 192 nodes led by Dubai, Doha, Shanghai Pudong, Istanbul and Incheon, hold 84.6 percent against 77.6 percent

of emissions themselves (Extended Data Fig. 8 and Extended Data Table 7). An instrument covering those nodes would cover 85 percent of the connectivity-driven total.

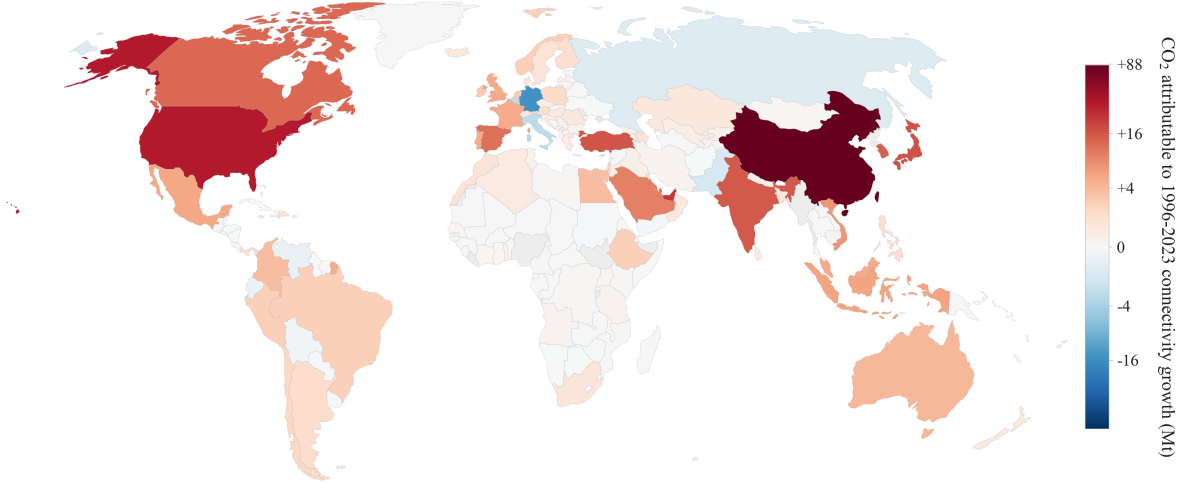


Figure 2: **Aviation CO₂ attributed to post-1996 connectivity growth, 2023 (Mt).** White denotes zero; the attribution applies the Feyrer-instrument elasticity to each country’s connectivity change.

Table 4: Country-level attribution and social cost of connectivity-driven aviation CO₂, 2023

	$\Delta \ln \text{GACI}$	Attributed CO ₂ (Mt)		Social cost (\$bn per year)		
		Total	International	SCC \$51	SCC \$185	SCC \$190
China	0.33	85.9	16.4	4.4	15.9	16.3
United States	0.04	34.5	12.5	1.8	6.4	6.6
United Arab Emirates	0.57	24.5	24.3	1.2	4.5	4.6
Japan	0.19	16.9	9.8	0.9	3.1	3.2
Turkey	0.55	16.5	13.3	0.8	3.0	3.1
India	0.19	15.4	7.2	0.8	2.8	2.9
Korea	0.71	14.1	12.7	0.7	2.6	2.7
Canada	0.22	12.7	8.1	0.6	2.4	2.4
Qatar	0.80	12.2	12.2	0.6	2.3	2.3
Spain	0.14	11.4	9.0	0.6	2.1	2.2
Germany	-0.09	-14.7	-13.5	-0.7	-2.7	-2.8
World		348.7	206.0	17.8	64.5	66.3

Notes: Attributed CO₂ applies the Feyrer-instrument elasticities (5.347 total, 5.192 international) to each country’s change in log connectivity over its observed span within 1996–2023: attributed share = $1 - \exp(-\beta \Delta \ln \text{GACI}_c)$, applied to 2023 bunker emissions (Methods). Social cost values one year’s attributable emissions at \$51, \$185 and \$190 per tonne of CO₂ (2020 USD; US Interagency Working Group, Rennert et al. 2022, US EPA 2023). Rows show the ten largest positive countries, Germany (largest negative) and the world total (184 countries), which equals 41.1 percent of 2023 aviation CO₂ (848 Mt). Country shares are extreme for fast growers because the elasticity exceeds one; the figures are partial-equilibrium accounting.

Only fuel switching materially lowers the level

Because the efficiency margin offsets 6 percent of the scale margin, the level of emissions can be materially lowered only by changing the fuel burned. Projected to 2050 with connectivity growing at its 2010–2019 pace (0.74 percent per year on average across countries; 0.45 percent in a low case), the mature-network elasticity of 2.9, and ReFuelEU blending milestones from 2 percent in 2025 to 70 percent in 2050 at life-cycle savings of 50, 65 and 80 percent (Fig. 3; Methods), world aviation CO₂ without fuel switching reaches 1,528 Mt in 2050 (1,213 Mt in the low-growth case), 1.8 times the 2023 level of 848 Mt. The full blending path with a 65 percent saving returns 2050 emissions to the 2023 level (833 Mt), and the 80 percent saving takes them to 672 Mt; no path reaches the 499 Mt that would remain after removing the emissions attributed to past connectivity growth. Holding 2023 traffic fixed, as in a pure accounting exercise, understates 2050 emissions under the same blending paths by 300 to 450 Mt.

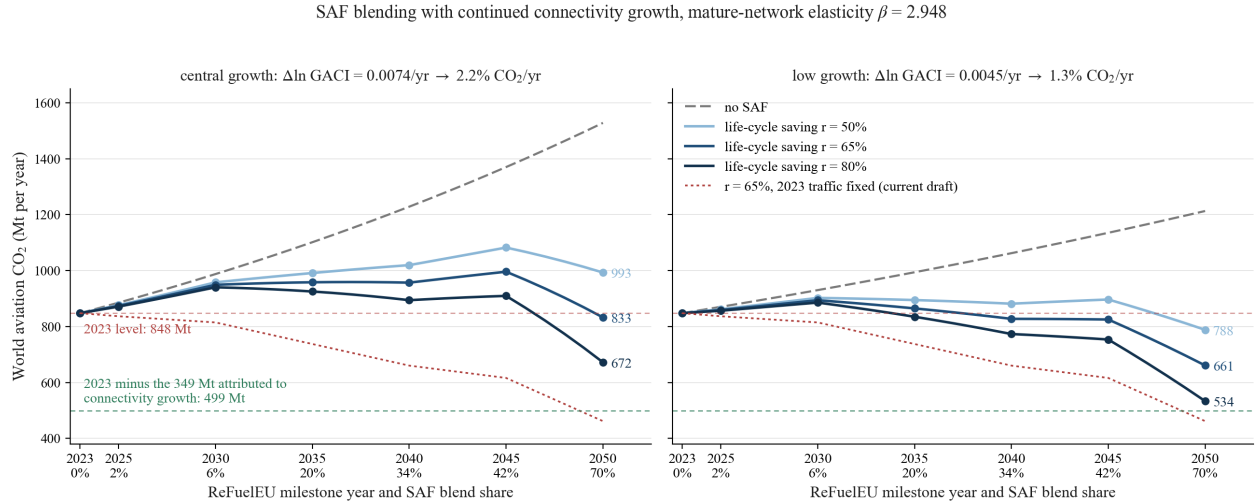


Figure 3: **World aviation CO₂ to 2050 under continued connectivity growth and ReFuelEU-style SAF blending.** Left, connectivity growth at its 2010–2019 pace; right, a low-growth case. Solid lines apply life-cycle savings of 50, 65 and 80 percent along the ReFuelEU blending milestones (shares on the axis); the dashed grey line is the no-SAF path, the dotted red line the 65 percent path with 2023 traffic held fixed, and the horizontal lines mark the 2023 level (848 Mt) and the level net of the emissions attributed to 1996–2023 connectivity growth (499 Mt). Traffic growth is converted to emissions with the mature-network elasticity (2.9).

Discussion [Jinwoo]

Our results provide global causal evidence that air connectivity is a system-level driver of aviation CO₂, with effects that are strongest during network take-off and extend across national borders. Five implications follow. First, connectivity raises emissions through flight frequency. Because GACI combines topological and volumetric attributes, its elasticity captures system-level development rather than a single route. A one percent increase in GACI raises seat-kilometres by 5.64 percent and CO₂ by 5.35 percent. Additional flights carry the response; changes in aircraft gauge, stage length and emissions intensity are not individually significant under country-clustered inference. The 0.29 percent intensity decline offsets only 6 percent of the scale response; full offset would require about 6 percent. Aviation decarbonisation pathways often treat traffic growth as exogenous, whereas we identify network development as an important source [TODO cite: aviation decarbonisation pathways]. Within the scenarios considered, sustained fuel switching materially lowers total emissions; the observed efficiency gains do not. Policy should therefore track absolute CO₂ and traffic growth and combine connectivity expansion with lower-carbon fuels, propulsion technologies and demand-side measures.

Second, the global mean elasticity does not apply equally across countries. Identification is concentrated in the take-off stage of network development, and the response is largest in countries with low baseline connectivity. Continuous estimates suggest a declining, rather than vanishing, response. Earlier work finds that connectivity yields its largest economic gains in low-income, remote and weakly connected economies [TODO cite: companion GACI paper]. Marginal economic benefits and carbon consequences may therefore be greatest in the same places. These results do not justify restricting access; they support coupling network expansion with early provision of lower-carbon fuels, cleaner aircraft and supporting infrastructure.

Third, connectivity creates effects across borders. A one percent increase in contiguous neighbours' connectivity raises own-country aviation CO₂ by about 2 percent, mainly through international traffic. The effect follows land borders rather than a smooth distance gradient, indicating a regional network externality rather than a global spillover, and it operates through

neighbours’ connectivity rather than through neighbours’ emissions: spatial lag and spatial error models leave the own elasticity and the total effect unchanged. Separately, bunker conventions shift measured responsibility across countries by up to 3.5 percentage points of the world total, so recorded emissions may fall outside the country whose network change generated the traffic. The social-cost estimate is sizeable but depends on attribution and carbon valuation. Both wedges matter for CORSIA, whose monitoring is based on reported international-flight emissions and whose offsetting obligations follow covered State pairs rather than the regional causal incidence of network expansion [TODO cite: ICAO CORSIA]. National measures should therefore be complemented by regional coordination in airport investment, route development and low-carbon fuel supply.

Fourth, the airport results suggest where mitigation could be targeted. Legacy hubs have lower conditional elasticities, so the marginal response is not confined to mega-hubs. Nevertheless, 85 percent of attributed emissions is concentrated in the top 5 percent of airports, with emerging hubs among the largest contributors. These descriptive estimates identify high-leverage sites for low-carbon fuel supply and infrastructure, not sole sources of causation or responsibility; the response extends across countries and routes.

Fifth, these implications should be interpreted in light of several limitations. Three direct diagnostics are consistent with the exclusion restriction, but identification is concentrated in the network-expansion era and in lower-connectivity countries. The later-period elasticity is identified only after excluding the COVID-19 collapse years; correlated global cycles and sectoral change remain possible channels. Emissions are schedule-based and do not reflect realised load factors. Airport estimates are descriptive; the 35–41 percent attribution range and fuel-switching scenarios are partial-equilibrium calculations; and non-CO₂ climate effects are excluded. Policy assessments should therefore use development-stage-specific ranges rather than one global elasticity. Future work could separate GACI’s topological and volumetric components and use route-level causal designs to test whether new links, centrality gains or traffic concentration drive changes in demand, schedules and fleet assignment. Coupling these pathways with deployment scenarios for sustainable aviation fuels, synthetic e-fuels and alternative propulsion systems could

show when technological change offsets connectivity-induced growth in flights.

Methods

Flight-level aviation CO₂ emissions

We estimate flight-level CO₂ emissions from global scheduled flight records provided by the Official Airline Guide (OAG), covering January 1996 to June 2024, following the aviation emissions methodology of the European Monitoring and Evaluation Programme/European Environment Agency (EMEP/EEA) Air Pollutant Emission Inventory Guidebook (EEA, 2023) and using engine-specific operating parameters from the International Civil Aviation Organization (ICAO) Aircraft Engine Emissions Databank. For each physically operated flight leg f , total CO₂ emissions are separated into landing-and-take-off (LTO) and climb–cruise–descent (CCD) components:

$$E_{f,\text{Total}} = E_{f,\text{LTO}} + E_{f,\text{CCD}}. \quad (1)$$

The LTO component covers aircraft operations below 3,000 feet, while the CCD component covers climb, cruise and descent above 3,000 feet. In the EEA implementation used here, LTO emissions are calculated for five components: taxi-out, take-off, climb-out, approach and landing, and taxi-in. Multi-stop scheduled services are decomposed into their constituent operated legs before emissions are calculated, so that each physical leg contributes its own LTO and CCD emissions rather than being treated as a single end-to-end flight.

OAG equipment codes are matched to aircraft types and representative aircraft–engine configurations in the EEA aviation emissions database. Where an exact aircraft type is unavailable, we assign a representative type from the same aircraft family or series, or the closest operational substitute on the basis of aircraft size and operating characteristics. The resulting aircraft–engine mapping links each scheduled flight leg to the aircraft-specific parameters required for the LTO and CCD calculations.

For LTO emissions, we retain the five operating components separately. Let $j(f)$ denote the aircraft–engine configuration assigned to flight f , and let k index the five LTO components. CO₂ emissions for component k are calculated as

$$E_{f,k}^{\text{LTO}} = T_k FF_{j(f),k} NE_{j(f)} EF_{\text{CO}_2}, \quad E_{f,\text{LTO}} = \sum_{k \in \mathcal{K}_{\text{LTO}}} E_{f,k}^{\text{LTO}}. \quad (2)$$

where T_k is the standard time in mode, $FF_{j(f),k}$ is the fuel-flow rate per engine for the corresponding operating mode, $NE_{j(f)}$ is the number of engines, and EF_{CO_2} is the fuel-specific CO₂ emission factor. We use standard time-in-mode values of 19 minutes for taxi-out, 0.7 minutes for take-off, 2.2 minutes for climb-out, 4 minutes for approach and landing, and 7 minutes for taxi-in. Engine-specific fuel-flow rates are obtained from the International Civil Aviation Organization (ICAO) Aircraft Engine Emissions Databank. CO₂ emission factors are 3.15 kilograms of CO₂ per kilogram of Jet A and 3.10 kilograms of CO₂ per kilogram of aviation gasoline. Retaining the individual LTO components also allows departure- and arrival-side emissions to be distinguished in the country-level allocation described below.

CCD emissions are estimated from aircraft-specific reference values provided by the EEA at discrete flight distances. For each flight leg, great-circle distance between the origin and destination airports is converted to the corresponding EEA flight distance following Avogadro and Redondi (2024). When this distance d_f falls between two adjacent EEA reference distances d_1 and d_2 , CCD emissions are obtained by linear interpolation:

$$E_{f,\text{CCD}}(d_f) = E_{j(f),\text{CCD}}(d_1) + [E_{j(f),\text{CCD}}(d_2) - E_{j(f),\text{CCD}}(d_1)] \frac{d_f - d_1}{d_2 - d_1}. \quad (3)$$

This procedure preserves the aircraft-specific EEA fuel-burn relationships while allowing emissions to vary continuously with flight distance. The resulting inventory therefore provides standardized LTO- and CCD-specific CO₂ estimates for each scheduled flight leg, rather than attempting to reproduce realized trajectories or flight-specific meteorological and operational

conditions.

Allocation of emissions to countries

Constructing country-year emissions from the flight-level inventory requires an explicit rule for attributing emissions to national units. This is relatively straightforward for LTO emissions, because the underlying operations can be associated with the departure or arrival airport. The attribution of CCD emissions is less direct, particularly for international flights, because the en-route component does not have a unique country assignment. Country-level aviation emissions therefore depend in part on the accounting convention used to allocate this component. To separate locally anchored airport emissions from this accounting choice, and to assess the sensitivity of our empirical results to alternative treatments of international en-route emissions, we construct three country-level measures.

Because the five LTO components are retained separately in the flight-level inventory, we distinguish the emissions associated with the departure and arrival sides of each flight. For flight leg f , define

$$D_f = E_{f,\text{taxi-out}} + E_{f,\text{take-off}} + E_{f,\text{climb-out}},$$

$$A_f = E_{f,\text{approach/landing}} + E_{f,\text{taxi-in}}, \quad C_f = E_{f,\text{CCD}}.$$

Total emissions from the flight leg can therefore be written as

$$E_{f,\text{Total}} = D_f + A_f + C_f. \tag{4}$$

For a flight from airport A to airport B , D_f is assigned to the country containing the departure airport and A_f to the country containing the arrival airport. This links LTO emissions to the location of the corresponding airport operation. Flights are classified as domestic or international according to whether their origin and destination airports are located in the same

country. The three national measures differ only in their treatment of C_f .

The first is an LTO-only measure, which attributes to each country only the emissions associated with airport operations occurring at its airports. Let O_{ct} and I_{ct} denote the sets of flight legs departing from and arriving in country c in year t , respectively. We define

$$E_{ct}^{\text{LTO}} = \sum_{f \in O_{ct}} D_f + \sum_{f \in I_{ct}} A_f. \quad (5)$$

Because this measure excludes CCD emissions by construction, it is not intended to represent a complete national aviation-emissions inventory. Rather, it provides a geographically anchored measure of local aviation emissions that is independent of any convention for allocating the en-route component of international flights.

Our headline measure additionally assigns the full CCD component of each flight to its country of departure:

$$E_{ct}^{\text{dep}} = E_{ct}^{\text{LTO}} + \sum_{f \in O_{ct}} C_f. \quad (6)$$

We refer to this as the departure-based, or bunker, measure. At the flight level, this allocation mirrors the origin-side logic of international bunker accounting: en-route emissions are associated with the departing side of the flight, while LTO emissions remain assigned to the airports at which the corresponding operations occur. The measure therefore provides a complete allocation of flight emissions across countries and serves as the principal emissions outcome in the empirical analysis.

As a symmetric alternative, we construct a 50/50 measure that assigns one half of each flight's CCD emissions to the origin country and the other half to the destination country:

$$E_{ct}^{50/50} = E_{ct}^{\text{LTO}} + \frac{1}{2} \sum_{f \in O_{ct}} C_f + \frac{1}{2} \sum_{f \in I_{ct}} C_f. \quad (7)$$

This allocation does not privilege either endpoint in assigning international en-route emissions and therefore provides a transparent robustness check on the origin-based convention.

For domestic flights, the departure-based and 50/50 measures are identical because both endpoints belong to the same country. Differences between the two full-emissions measures therefore arise entirely from the allocation of CCD emissions on international flights. Importantly, both allocation rules preserve the same global emissions total:

$$\sum_c E_{ct}^{\text{dep}} = \sum_c E_{ct}^{50/50} = \sum_{f \in F_t} (D_f + A_f + C_f), \quad (8)$$

where F_t denotes all flight legs in year t . Thus, differences across the national measures reflect only how a fixed quantity of flight-level emissions is distributed across countries, rather than differences in the underlying emissions inventory.

Validation of the emissions inventory

The flight-level inventory is intended to provide globally consistent estimates over nearly three decades rather than to reproduce the realized conditions of every individual flight. Meteorological conditions, routing and air-traffic-control adjustments, airport-specific operations, and other flight-level variation are therefore not observed, and a small number of aircraft types are represented by closely matched aircraft–engine configurations when no exact EEA counterpart is available. Standardized time-in-mode and aircraft-performance parameters likewise represent typical rather than realized operations. We therefore assess whether these approximations materially affect the dimensions of the inventory used in our analysis: the scale of emissions, their geographic distribution, and their LTO–CCD composition.

The scale and evolution of global aviation emissions are closely reproduced against two references constructed from fundamentally different sources. The International Air Transport Association (IATA) provides industry-level global CO₂ estimates based on airline operational and industry statistics, whereas the Organisation for Economic Co-operation and Development (OECD) constructs a bottom-up inventory from individual flight activity and engineering-based emissions calculations (IATA, 2018, 2019, 2024; Clarke et al., 2022). The convergence of these distinct approaches is informative because it does not arise from alternative implementations of

the same underlying model. Over 2004–2023, our estimates have a weighted absolute percentage error (WAPE) of 3.04% relative to IATA and an aggregate bias of -1.98% ; against the OECD inventory over 2013–2023, the corresponding values are 3.58% and $+3.40\%$ (Table 5).

IATA and OECD provide global coverage but remain estimated emissions series. We therefore complement them with carrier-reported fuel consumption from the United States (U.S.) Bureau of Transportation Statistics (BTS), which is geographically narrower but more directly measures the physical quantity underlying CO₂ emissions. Over 1996–2023, our estimated U.S.-carrier fuel consumption has a WAPE of 3.84% relative to BTS and an aggregate bias of $+0.76\%$. Together, the three sources test the inventory from complementary perspectives: global industry estimates, an independent flight-level inventory, and reported fuel use.

We next examine the cross-country distribution of emissions, which is directly relevant to the country-year empirical design. After applying a consistent departure-country allocation rule, the mean annual log-scale Pearson correlation with the OECD territory-based series is 0.999 over 2013–2018, with a mean total variation distance (TVD) of 1.05%. From 2019, when the OECD activity data shift from scheduled-flight records to realized flights observed through Automatic Dependent Surveillance–Broadcast (ADS-B), agreement remains strong but discrepancies increase: over 2019–2023, the mean correlation is 0.955 and mean TVD is 6.04%. Matched countries account for more than 99% of emissions in our inventory in every year. The post-2019 divergence is therefore consistent with differences between scheduled and realized flight activity rather than a broad disagreement in the geographic distribution of emissions.

Finally, the estimated flight-phase composition is also consistent with independent evidence. LTO operations account for 12.41% of total CO₂ emissions over 1996–2023. Quadros et al. (2022) report global LTO fuel-burn shares of 10.8–13.5% under ICAO reference LTO-cycle assumptions. Because CO₂ emissions from conventional aviation fuel are proportional to fuel burn, these values provide a directly comparable reference, and our estimate falls within the reported range.

Table 5: External validation of aviation emissions estimates

Benchmark	Comparison	Period	WAPE	Aggregate bias
IATA	Global CO ₂ emissions	2004–2023	3.04%	−1.98%
OECD	Global CO ₂ emissions	2013–2023	3.58%	+3.40%
BTS	U.S. carrier fuel consumption	1996–2023	3.84%	+0.76%

Notes: WAPE is defined as $\sum_t |\hat{E}_t - E_t^B| / \sum_t E_t^B$. Aggregate bias is defined as $\sum_t (\hat{E}_t - E_t^B) / \sum_t E_t^B$; positive values indicate that our estimate exceeds the corresponding benchmark.

Global Air Connectivity Index

We measure air connectivity using the Global Air Connectivity Index (GACI), following Cheung et al. (2020). GACI is a composite measure of an airport’s position in the global air transport network, combining structural and volumetric dimensions of connectivity, including direct and indirect connectivity, capacity-based link intensity and regional importance. Because the evolution of connectivity is central to our analysis, we reconstruct the worldwide airport network separately for each year from OAG route-segment schedules. Airports therefore enter and leave the network as scheduled service begins or ceases, and connectivity can vary continuously with changes in routes, frequencies and capacity.

Let V_t denote the set of airports with active scheduled service in year t . Airports are represented as nodes and scheduled connections as undirected links, with $a_{ijt} = 1$ if airports i and j are directly connected and zero otherwise. Let q_{ijt} denote annual scheduled seat capacity on the link, aggregated over both directions. Airport strength and degree are

$$s_{it} = \sum_{j \in V_t} q_{ijt}, \quad d_{it} = \sum_{j \in V_t} a_{ijt}. \quad (9)$$

For each active link $\{i, j\}$, connection intensity is defined as

$$w_{ijt} = \sqrt{\frac{s_{it}s_{jt}}{d_{it}d_{jt}}}, \quad a_{ijt} = 1, \quad (10)$$

and $w_{ijt} = 0$ otherwise. This measure is high when both endpoint airports carry substantial scheduled capacity per connection and provides the common measure of connection strength used in the weighted components of GACI.

GACI combines five complementary indicators of airport network position.

Degree measures the number of airports directly connected to airport i :

$$d_{it} = \sum_{j \in V_t} a_{ijt}. \quad (11)$$

It captures the direct reach of an airport within the global network.

Flow betweenness captures the extent to which airport i lies on potential transfer flows between other origin–destination pairs:

$$F_b(i, t) = \frac{1}{N_{B,t}} \sum_{\substack{j, g \in V_t \\ j \neq g, j \neq i, g \neq i}} \tau_{jgt}(i), \quad N_{B,t} = \frac{N_t(N_t - 1)}{2}, \quad (12)$$

where $N_t = |V_t|$ and $\tau_{jgt}(i)$ denotes modelled flow through airport i between airports j and g under an electrical-circuit representation of the weighted network, with link conductance proportional to w_{ijt} . The measure therefore captures an airport's importance as an intermediary in the network rather than only its direct connections.

Closeness centrality measures the topological accessibility of airport i :

$$C_c(i, t) = \frac{N_t - 1}{\sum_{\substack{j \in V_t \\ j \neq i}} \text{Dist}_t(i, j)}, \quad (13)$$

where $\text{Dist}_t(i, j)$ is the shortest-path length between airports i and j , measured in flight legs. Closeness therefore reflects how few connections are required to reach the rest of the network rather than geographic distance, flight distance or travel time.

Eigenvector centrality assigns greater weight to connections with airports that are themselves highly connected:

$$C_E(i, t) = \frac{1}{\lambda_t} \sum_{j \in V_t} w_{ijt} C_E(j, t), \quad (14)$$

where λ_t is the largest eigenvalue of the weighted adjacency matrix $W_t = [w_{ijt}]$. Connections

to more central airports therefore contribute more strongly to an airport's network importance.

Finally, regional importance captures the strength of an airport's connections within its own OAG world region. Let $R(i)$ denote the region containing airport i , and define

$$V_{it}^R = \{j \in V_t : a_{ijt} = 1, R(j) = R(i)\}. \quad (15)$$

Regional importance is

$$C_R(i, t) = \frac{1}{|V_{it}^R|} \sum_{j \in V_{it}^R} w_{ijt}. \quad (16)$$

This component measures the strength of an airport's position within its regional aviation network. Taken together, degree, closeness and eigenvector centrality primarily characterize network topology, whereas flow betweenness and regional importance additionally incorporate the intensity of airport connections. The five measures therefore capture distinct but complementary dimensions of connectivity.

Because the five indicators differ in scale, they are standardized following Cheung et al. (2020) and combined using principal component analysis (PCA). Let $\tilde{d}_{it}$, $\tilde{f}_{it}$, $\tilde{c}_{it}$, $\tilde{e}_{it}$ and $\tilde{r}_{it}$ denote the standardized degree, flow-betweenness, closeness, eigenvector-centrality and regional-importance measures. Airport-level GACI is defined by the first principal component:

$$\text{GACI}_{it} = \begin{bmatrix} \tilde{d}_{it} & \tilde{f}_{it} & \tilde{c}_{it} & \tilde{e}_{it} & \tilde{r}_{it} \end{bmatrix} \boldsymbol{\alpha}_t, \quad (17)$$

where $\boldsymbol{\alpha}_t$ is the first-principal-component loading vector estimated for the year- t network. Across 1996–2023, the loadings are uniformly positive and highly stable, and the first component explains 63–77% of annual variation in the five standardized indicators. A pooled PCA estimated over all airport-year observations yields an index correlated at 0.98 with the year-specific construction, supporting the intertemporal comparability of the index.

Our empirical analysis is conducted primarily at the country level. Let A_{ct} denote the set of airports with active scheduled service in country c in year t . The headline national measure is

the capacity-weighted mean (cwm) of airport GACI:

$$\text{GACI}_{ct}^{\text{cwm}} = \frac{\sum_{i \in A_{ct}} s_{it} \text{GACI}_{it}}{\sum_{i \in A_{ct}} s_{it}}. \quad (18)$$

This measure summarizes the connectivity of the national airport system while assigning greater weight to airports that account for larger shares of scheduled capacity.

We additionally consider three alternative aggregations. The maximum

$$\text{GACI}_{ct}^{\text{max}} = \max_{i \in A_{ct}} \text{GACI}_{it} \quad (19)$$

captures the connectivity of the country's leading gateway. The sum

$$\text{GACI}_{ct}^{\text{sum}} = \sum_{i \in A_{ct}} \text{GACI}_{it} \quad (20)$$

captures aggregate airport-system connectivity, whereas the unweighted mean

$$\text{GACI}_{ct}^{\text{mean}} = \frac{1}{|A_{ct}|} \sum_{i \in A_{ct}} \text{GACI}_{it} \quad (21)$$

assigns equal weight to all active airports. The capacity-weighted mean is our headline measure; the alternative aggregations assess the robustness of the results to different representations of national network structure.

GACI is particularly suited to the present analysis because aviation connectivity evolves continuously rather than only through discrete airport or route openings. Changes in flight frequency and seat capacity, shifts in the connectivity of partner airports, and changes in indirect access can alter a country's network position even when its set of direct routes is unchanged. GACI therefore captures both extensive-margin changes in network coverage and intensive-margin changes in the strength and importance of existing connections.

The estimation sample covers 184 countries over 1996–2023, comprising 4,634 country-year observations. Of the 6,356 airports represented in the emissions inventory, 5,354 are nodes in

the connectivity network; airports not matched to the GACI network account for only 0.1% of departure-based emissions, and airports that cannot be assigned to a country (0.02% of emissions) are excluded from the country panel.

Estimation [Junya]

The estimating equation is

$$\ln \text{CO}_{2ct} = \beta \ln \text{GACI}_{ct} + \gamma \ln \text{pop}_{ct} + \delta \ln \text{seaMA}_{ct} + \alpha_c + \tau_t + \varepsilon_{ct}, \quad (22)$$

estimated by two-stage least squares with country and year fixed effects and standard errors clustered by country (184 clusters); all first-stage statistics are cluster-robust. Heteroskedasticity-robust standard errors, which are about one third as large, are reported for comparison in Supplementary Table 1, and two-way clustering by country and year gives the same standard error as country clustering.

Decomposition on the accounting identity [Junya]

Because emissions are computed from scheduled flights, national CO_2 satisfies

$$\text{CO}_{2ct} = F_{ct} \times \frac{S_{ct}}{F_{ct}} \times \frac{K_{ct}}{S_{ct}} \times \frac{\text{CO}_{2ct}}{K_{ct}}, \quad (23)$$

where F is departing flights, S departing seats and K departing seat-kilometres, so that seats per flight is aircraft gauge, kilometres per seat is the seat-weighted average stage length, and CO_2 per seat-km is carbon intensity. Load factor multiplies scheduled capacity by a constant and cancels. Estimating equation 22 for the log of each factor on the identical sample gives

$$\beta_{\text{CO}_2} = \beta_{\text{flights}} + \beta_{\text{gauge}} + \beta_{\text{stage}} + \beta_{\text{intensity}}, \quad (24)$$

which holds exactly because two-stage least squares is linear in the outcome. The scale subtotal is the seat-kilometre elasticity. The same identity is applied separately to international and domestic departures and within baseline income and connectivity terciles.

Airport-level design and identification pilot [Junya]

Airport regressions include airport and country-by-year fixed effects, with standard errors clustered by airport, and are estimated on subsamples defined by 1996 characteristics: global rank by seat-kilometres, terciles of baseline connectivity and traffic, baseline international share, region, and the host country's income tercile. Each country's largest airport cannot be estimated as a separate subsample under country-by-year effects and enters through a pooled interaction. Because airport GACI contains a capacity term, we also use the topology-only components of the index (eigenvector, closeness, betweenness, degree) as regressors. The identification pilot interacts the world seat-capacity index with each airport's 1996 air market access, the population-weighted inverse distance to all foreign countries' aviation centroids; within countries this term has a mean standard deviation of 0.04 and no first stage (partial F of 0.4 to 0.7 with and without size-by-cycle controls; Supplementary Table 2). A heritage-proximity instrument piloted earlier failed the size-by-cycle test (59.5 to 2.3). Airport-level estimates are therefore within-country descriptive evidence. The concentration analysis ranks airports by attributed emissions, computed as in the country attribution with the country elasticity and, as sensitivity, with the airport within-country elasticity and with separate elasticities for the 1996 top five percent and other airports, and compares the shares held by the top one, five, ten and 25 percent of airports with the same shares for 2023 emissions.

Instruments and validity [Junya]

Connectivity is instrumented with a Feyrer-type shifter that interacts world aviation technology a_t with the country's air market access computed on fixed 1996 geography, while the sea-distance market-access channel is controlled directly. The identifying variation is the differential exposure

of countries to the global growth of aviation implied by their geography, holding the surface-shipping alternative fixed; the first-stage Kleibergen–Paap F is 18 (154 with heteroskedasticity-robust errors). Because any shifter of the form (global cycle $\times$ baseline characteristic) may proxy (global cycle $\times$ country size), we re-estimate each first stage adding interactions of the global cycle with baseline population, GDP, and airport capacity. The primary instrument survives (on the 4,121 country-years of countries observed in 1996, for which baseline size is defined, the heteroskedasticity-robust F declines only from 120.1 to 105.8); a tourism-heritage instrument does not (13.8 to below 1) and is used only as a lens on the composition margin, its compliers being hub-quality changes with a post-2010 first stage. An aviation-accident instrument (lagged fatal accidents and accident deaths from the Aviation Safety Network) is too weak to use on its own (heteroskedasticity-robust first-stage F of 1 to 4; 2SLS 5.6 with a standard error of 2.2 for lagged fatal accidents), but the Hansen test does not reject its joint validity with the Feyrer instrument ($p = 0.81$ and 0.95), and the jointly instrumented elasticity is 5.1 (Supplementary Table 3, heteroskedasticity-robust). Plausibly-exogenous bounds indicate that a direct effect of the instrument would need to account for 85 percent of the reduced form (42 percent for intensity) to overturn the results; quintile reduced forms are reported in Extended Data Fig. 9.

Exclusion-restriction diagnostics [Junya]

The exclusion restriction requires that the instrument affects aviation CO₂ only through air connectivity. Three tests address it directly, and all three pass (Extended Data Table 11, Panels A, C and D; Extended Data Fig. 12). First, non-aviation emissions do not respond to the instrument: the reduced form on total territorial fossil CO₂ excluding domestic aviation is 0.05 (s.e. 0.12) against 0.68 (s.e. 0.15) for aviation CO₂, so a general development channel that would raise all emissions is rejected. Second, the effect runs through air geography rather than geography in general: when the aviation-technology cycle is interacted with 1996 sea market access instead of air market access and both interactions are entered, the sea term has a reduced form of -0.19 (s.e. 0.26) while the air term retains 0.82 (s.e. 0.21). Third, where the instrument

cannot move connectivity it does not move emissions: in the top tercile of baseline connectivity the first stage is absent ($F = 0.2$) and the reduced form on CO₂ is 0.09 (s.e. 0.19), against 1.97 and 0.41 in the lower terciles.

Six further checks are reported in Extended Data Table 11 and Supplementary Table 1: development-channel controls (the elasticity stays between 4.4 and 4.6), alternative technology series and distance-decay exponents (5.2 to 5.8), leads of the instrument (uninformative, because the technology index is a smooth trend), the Hansen test against the tourism-heritage instrument (rejects at 5 percent, as expected for an instrument with different compliers that fails the stress test), a horse race against rival global cycles, and the comparison of heteroskedasticity-robust and clustered errors. Two of these leave caveats that we state rather than resolve: coal and cement CO₂ rise with the instrument (0.90 and 0.67), indicating some recomposition of non-aviation emissions in countries whose air access improved, and the aviation cycle cannot be separated statistically from world GDP and trade cycles interacted with the same geography (time-series correlations of 0.82 and 0.84), so the aviation-specific content of the identifying variation rests on the air-versus-sea contrast of the second test rather than on the cycle itself. Alternative instrument constructions that would avoid these caveats, the air-minus-sea advantage interacted with the cycle and the air interaction with the sea interaction as a control, have weak first stages (country-clustered F of 1.0 and 5.6).

Spillover design [Junya]

Neighbour connectivity is the leave-out average of other countries' log GACI under a weighting matrix, instrumented by the identically weighted average of their Feyrer shifters, with own connectivity instrumented by the own shifter in the same regression. Weightings are land contiguity from Natural Earth administrative boundaries (38 sample countries have no land neighbour and receive a zero exposure with an indicator), the five nearest countries and distance bands by aviation-centroid distance, exponential kernels $\exp(-d/\lambda)$, leave-out means within UN sub-regions and within aviation blocs, and the all-country inverse-distance weighting of the earlier

draft. Conditional first-stage strength is the Sanderson–Windmeijer F for each endogenous regressor. Under the inverse-distance weighting the two shifters are nearly collinear within country and the joint first stage fails (F about 5); that specification is retained only with the own shifter in reduced form. Regional trends are absorbed by sub-region-by-year fixed effects in the robustness specifications. Because clustering by country weakens the joint first stage, inference on the neighbour coefficient uses Anderson–Rubin sets obtained by inverting, over a grid of null values, the cluster-robust test that the neighbour instrument has no explanatory power for the outcome net of the null; in the joint specification own connectivity remains instrumented under each null (subset Anderson–Rubin). The permutation placebo relabels countries at random in the weight matrix, rebuilds the exposures and re-estimates the specification 500 times; the p -value is the share of draws whose t -statistic on the neighbour term exceeds the actual one in absolute value.

Spatial models [Junya]

The spatial specifications in Table 3 extend equation 22 with, respectively, the neighbours' average log connectivity $W \ln \text{GACI}_{ct}$ (SLX), the neighbours' average log emissions $\rho W \ln \text{CO}_{2ct}$ (SAR), a spatially autoregressive disturbance $u_{ct} = \lambda W u_{ct} + \varepsilon_{ct}$ (SEM), the spatial lag together with the neighbours' connectivity (SDM), and the neighbours' connectivity together with the spatial error (SDEM). W is the row-normalised land-contiguity matrix, applied within each year to the countries in the sample that year, so that the stacked matrix is block-diagonal by year; countries without a land neighbour have a zero row. Panel A estimates the models with country and year fixed effects on the within-transformed data, by least squares for SLX and by maximum likelihood for the spatial-lag and spatial-error models, with the log-determinant summed over the yearly blocks. Panel B instruments own connectivity with the Feyrer shifter, neighbours' connectivity with the identically weighted average of the shifter, and the spatial lag of the outcome with the first and second spatial lags of the shifter and of the exogenous controls, following Kelejian and Prucha's generalised spatial two-stage least squares; the spatial-error parameter is estimated by their moments estimator from the 2SLS residuals, after which

the equation is re-estimated on the spatially filtered variables. Direct, indirect and total effects of connectivity are the averages, over the yearly blocks, of the diagonal, the off-diagonal row sums and the row sums of $(I - \rho W)^{-1}(\beta I + \theta W)$, with standard errors from 100 draws of the coefficient vector from its estimated covariance. Standard errors are clustered by country in Panel B. Extended Data Table 10 repeats Panel B with inverse-distance weights and for international CO₂ and carbon intensity.

Mediation [Junya]

The instrumental-variable decomposition instruments the treatment with the Feyrer shifter in both the mediator and outcome equations and combines the paths by the product method with delta-method standard errors; the Imai, Keele and Yamamoto algorithm is implemented by quasi-Bayesian simulation on least-squares models with country and year indicators. Each mediator is analysed separately and the mediators overlap, so shares do not sum.

Attribution and fuel-switching scenarios [Ray and Lisa]

The share of 2023 emissions attributed to post-1996 connectivity growth is $1 - \exp(-\beta \Delta \ln \text{GACI}_c)$, applied to 2023 levels; because β far exceeds one, country-level shares are extreme for fast-growing countries and the world total (41.1 percent) is the more conservative summary. For the 34 countries entering the panel after 1996 and the 18 whose series end before 2023, the change is taken between the first and last observed years. The airport-level attribution applies the same formula to each airport's change in log GACI between 1996, or its first year in the network, and 2023; its sum (346 Mt) is close to the country total because the elasticity is applied to the same emissions base. Fuel-switching scenarios project world emissions as $E(t) = E_{2023} \exp[\beta g(t - 2023)](1 - s_t r)$, where g is the average annual change in log connectivity across countries over 2010–2019 (0.0074; 0.0045 in the low case), β the mature-network elasticity (2.9; Extended Data Table 1, Panel C), s_t the ReFuelEU blending share interpolated between the 2025–2050 milestones (2 to 70 percent) and r the life-cycle saving of sustainable

aviation fuel (50, 65 and 80 percent). Because β is a reduced-form emissions elasticity it already contains the efficiency margin, so no separate intensity trend is applied; ReFuelEU is used as a benchmark blending path rather than as a forecast of global mandates, and the calculations are partial-equilibrium accounting. Attributed emissions are monetised at three social-cost-of-carbon estimates, in 2020 USD per tonne of CO₂: the US Interagency Working Group interim value (\$51, 3 percent discount rate), the estimate of Rennert et al. (2022) (\$185, 2 percent near-term discount rate), and the US EPA (2023) central value (\$190, 2 percent Ramsey discounting) (Interagency Working Group on Social Cost of Greenhouse Gases, 2021; Rennert et al., 2022; US Environmental Protection Agency, 2023). The valuation applies to one year’s attributable emissions and recurs annually while the connectivity-driven traffic persists.

Data availability

[TODO: OAG licensing statement; GACI panel availability; replication archive.]

Code availability

[TODO: replication code archive.]

Extended Data

Extended Data Table 1: Temporal split of the connectivity elasticity

	Bunker CO ₂ (1)	Intl. CO ₂ (2)	Seat-km (3)	Intensity (4)
<i>Panel A. Full sample, 1996–2023</i>				
ln GACI (cwm)	5.347*** (1.076)	5.192*** (1.043)	5.641*** (1.194)	-0.294 (0.286)
KP F	18.4	18.4	18.4	18.4
N	4,634	4,634	4,634	4,634
<i>Panel B. Network-expansion era, 1996–2007</i>				
ln GACI (cwm)	5.609*** (1.092)	5.578*** (1.082)	4.754*** (0.993)	0.854** (0.371)
KP F	25.4	25.4	25.4	25.4
N	1,883	1,883	1,883	1,883
<i>Panel C. Mature-network era, 2010–2023 (excl. 2020–2021)</i>				
ln GACI (cwm)	2.948** (1.199)	2.904*** (1.095)	3.862*** (1.199)	-0.914* (0.494)
KP F	10.7	10.7	10.7	10.7
N	2,065	2,065	2,065	2,065

Notes: 2SLS with the Feyrer instrument; specifications as in Table 1. Panels B and C split the sample at its midpoint (2010) and exclude the crisis years: Panel B ends before the global financial crisis (2008–2009) and Panel C drops the COVID-19 collapse (2020–2021). The unrestricted 2010–2023 split (not shown) has a weak first stage (KP $F = 3.2$), driven by the COVID years, whose exclusion in Panel C restores F to 10.7; its estimates appear in grey in Extended Data Fig. 1. Standard errors clustered by country in parentheses; Kleibergen–Paap F statistics are cluster-robust. ***, **, *: significance at 1, 5, and 10 percent.

Extended Data Table 2: Allocation of emissions to countries: bunker, territorial (LTO) and 50/50 rules

	Bunker CO ₂ (1)	LTO CO ₂ (2)	50/50 CO ₂ (3)
<i>Panel A. OLS</i>			
ln GACI (cwm)	3.388*** (0.364)	2.916*** (0.366)	3.365*** (0.362)
<i>N</i>	4,634	4,634	4,634
<i>Panel B. 2SLS, Feyrer instrument, capacity-weighted mean</i>			
ln GACI (cwm)	5.347*** (1.076)	5.701*** (1.182)	5.300*** (1.069)
KP <i>F</i>	18.4	18.4	18.4
<i>N</i>	4,634	4,634	4,634
<i>Panel C. 2SLS, GACI sum</i>			
ln GACI (sum)	2.153*** (0.524)	2.296*** (0.502)	2.134*** (0.517)
KP <i>F</i>	14.0	14.0	14.0
<i>N</i>	4,634	4,634	4,634
<i>Panel D. 2SLS, GACI max</i>			
ln GACI (max)	4.548*** (0.841)	4.850*** (0.899)	4.508*** (0.835)
KP <i>F</i>	21.0	21.0	21.0
<i>N</i>	4,634	4,634	4,634
<i>Panel E. 2SLS, GACI mean</i>			
ln GACI (mean)	5.562*** (1.389)	5.931*** (1.465)	5.514*** (1.384)
KP <i>F</i>	29.3	29.3	29.3
<i>N</i>	4,634	4,634	4,634

Notes: Specifications as in Table 1. Column (1) repeats the headline bunker measure (all cruise plus departure-side LTO assigned to the departure country); column (2) uses landing-and-take-off emissions only (territorial rule); column (3) splits cruise emissions equally between the endpoint countries. The three rules give elasticities within 0.3 of one another under every aggregation of airport connectivity. Standard errors clustered by country in parentheses; the Kleibergen–Paap *F* is cluster-robust. ***, **, *: significance at 1, 5, and 10 percent.

Extended Data Table 3: Heterogeneity of the connectivity elasticity

	Coefficient	s.e.	KP F	N
<i>Panel A. Total CO₂ by baseline income tercile</i>				
Low income	9.427***	(2.652)	4.8	1,596
Middle income	6.118***	(2.115)	5.7	1,547
High income	1.886	(2.437)	1.7	1,491
Pooled interaction, top tercile $\times$ ln GACI	-3.257***	(0.565)	8.4	4,634
<i>Panel B. Total CO₂ by baseline connectivity tercile</i>				
Low connectivity	7.481***	(1.695)	16.4	1,453
Middle connectivity	5.068**	(2.497)	3.7	1,526
High connectivity (weak first stage)	4.035	(5.735)	0.2	1,655
Pooled interaction, top tercile $\times$ ln GACI	-1.997***	(0.676)	8.7	4,634
<i>Panel C. Total CO₂ by region</i>				
Europe	-0.466	(1.569)	6.6	1,190
Asia-Pacific	5.053***	(1.621)	10.1	1,110
Africa	4.542*	(2.617)	4.9	1,230
Latin America	2.926	(1.945)	1.7	683
Middle East	not identified (KP $F < 2$)			
North America	not identified (KP $F < 2$)			
<i>Panel D. Intensity by baseline income tercile</i>				
Low income	-0.725	(0.701)	4.8	1,596
Middle income	-0.372	(0.536)	5.7	1,547
High income	1.054	(1.501)	1.7	1,491

Notes: Split-sample 2SLS with the Feyrer instrument on log total bunker CO₂ (Panels A–C) and log CO₂ per seat-km (Panel D); terciles are formed on 1996 values. All specifications include country and year fixed effects and control for log population and log sea market access. The Middle East and North America cells have too little instrument variation for inference. Standard errors clustered by country in parentheses. ***, **, *: significance at 1, 5, and 10 percent.

Extended Data Table 4: Causal mediation of the connectivity effect on aviation CO₂

<i>Panel A. Instrumental-variable product of coefficients (2SLS scale; total effect 5.347)</i>					
Mediator	<i>a</i> : GACI → <i>M</i>	<i>b</i> : <i>M</i> → CO ₂	ACME (<i>a</i> × <i>b</i>)	Sobel <i>p</i>	Share mediated
Flight frequency	6.760	0.667	4.508	0.000	84.3%
Seat-kilometres	5.641	0.801	4.521	0.000	84.5%
International share	0.019	-1.605	-0.031	0.868	-0.6%
<i>Panel B. Imai, Keele and Yamamoto (2010) mediation (OLS scale; total effect 3.582)</i>					
Mediator	ACME	Direct effect	Total effect		Share mediated
Flight frequency	1.766	1.816	3.582		49.3%
Seat-kilometres	3.424	0.159	3.583		95.6%
International share	-0.077	3.657	3.580		-2.1%

Notes: Outcome is log total bunker CO₂; the treatment is log GACI (capacity-weighted mean). Panel A instruments the treatment with the Feyrer shifter in each equation (standard errors clustered by country) and combines the paths by the product method with delta-method (Sobel) standard errors. Panel B implements the algorithm of Imai, Keele and Yamamoto (2010) by quasi-Bayesian simulation (200 draws) on least-squares models with country and year indicator variables. Each row is a separate single-mediator analysis and the mediators overlap, so shares do not sum across rows. All models control for log population and log sea market access. Sequential ignorability of the mediator is assumed conditional on the fixed effects.

Extended Data Table 5: Decomposition of the CO₂ elasticity by group and segment

	Total CO ₂	Flights	Seats/flight	Km/flight	CO ₂ /seat-km	KP <i>F</i>	<i>N</i>
<i>Panel A. Baseline income tercile (total CO₂ identity)</i>							
Low income	+9.43*** (2.65)	+8.48*** (2.66)	+0.61 (0.93)	+1.07 (1.03)	-0.73 (0.70)	4.8	1,596
Middle income	+6.12*** (2.12)	+8.57*** (3.26)	-0.42 (0.89)	-1.66 (1.26)	-0.37 (0.54)	5.7	1,547
High income	+1.89 (2.44)	+7.96 (5.99)	-2.98 (2.86)	-4.15 (4.40)	+1.05 (1.50)	1.7	1,491
<i>Panel B. Baseline connectivity tercile</i>							
Low connectivity	+7.48*** (1.69)	+8.61*** (2.14)	-0.08 (0.59)	-0.31 (0.56)	-0.73 (0.45)	16.4	1,453
Middle connectivity	+5.07** (2.50)	+6.00** (2.74)	-0.11 (0.84)	-0.78 (1.77)	-0.05 (0.71)	3.7	1,526
High connectivity	+4.03 (5.74)	+18.93 (36.66)	-8.20 (18.81)	-8.52 (19.55)	+1.83 (4.88)	0.2	1,655
<i>Panel C. Segment identities</i>							
International traffic, all countries	+5.19*** (1.04)	+5.59*** (1.04)	+0.52 (0.37)	-0.46 (0.52)	-0.46 (0.30)	18.4	4,634
Domestic traffic, all countries	-0.75 (2.74)	+0.71 (2.76)	-0.15 (0.97)	-1.15** (0.55)	-0.16 (0.55)	11.4	3,558
International, low income	+9.85*** (2.87)	+8.45*** (2.58)	+0.74 (0.83)	+1.63* (0.98)	-0.98 (0.86)	4.8	1,596
International, high income	+0.59 (2.75)	+6.44 (4.15)	-0.97 (1.67)	-5.27 (4.74)	+0.39 (1.27)	1.7	1,491

2SLS elasticities to log GACI (Feyrer instrument; controls and fixed effects as in Table 2), estimated on the common sample of each block so that the four components sum to the total within each row. Panel C applies the identity separately to international and domestic departures; domestic rows exclude country-years without domestic traffic. The high-connectivity tercile has a weak first stage ($F = 0.2$) and is reported for completeness only. Standard errors clustered by country in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Extended Data Table 6: Airport-level elasticities within countries, by airport group

	log CO ₂	log seat-km	log CO ₂ /seat-km	Intl. share	Airport-years	Airports
<i>All airports</i>						
All	3.76*** (0.11)	4.23*** (0.12)	-0.50*** (0.04)	0.34*** (0.02)	94,582	4,881
<i>Panel A. Hub status (global rank by 1996 seat-km)</i>						
Top 1 percent	2.04*** (0.28)	2.23*** (0.37)	-0.19 (0.15)	0.46*** (0.12)	924	33
Others	3.81*** (0.11)	4.29*** (0.12)	-0.50*** (0.04)	0.34*** (0.02)	93,105	4,825
Top 5 percent	2.11*** (0.26)	2.25*** (0.27)	-0.14*** (0.05)	0.33*** (0.06)	5,474	200
Others	4.08*** (0.12)	4.60*** (0.13)	-0.55*** (0.05)	0.35*** (0.02)	87,426	4,614
<i>Panel B. Baseline connectivity tercile (1996 GACI)</i>						
Low	4.18*** (0.23)	4.82*** (0.25)	-0.58*** (0.10)	0.26*** (0.04)	20,639	1,504
Middle	5.84*** (0.23)	6.74*** (0.25)	-0.95*** (0.12)	0.49*** (0.05)	30,463	1,602
High	2.97*** (0.13)	3.23*** (0.14)	-0.31*** (0.05)	0.28*** (0.02)	39,921	1,614
<i>Panel C. Baseline traffic tercile (1996 seat-km)</i>						
Low	5.12*** (0.33)	5.67*** (0.42)	-0.54** (0.21)	0.24*** (0.07)	16,747	1,435
Middle	5.23*** (0.20)	6.03*** (0.21)	-0.85*** (0.09)	0.39*** (0.04)	32,408	1,638
High	3.11*** (0.12)	3.40*** (0.13)	-0.33*** (0.04)	0.32*** (0.02)	42,317	1,649
<i>Panel D. Orientation (baseline international share of seat-km)</i>						
Domestic only	4.26*** (0.13)	4.77*** (0.15)	-0.53*** (0.06)	0.30*** (0.02)	65,027	3,614
Below one half	2.91*** (0.23)	3.27*** (0.23)	-0.44*** (0.13)	0.44*** (0.04)	14,534	551
Above one half	4.72*** (0.29)	5.27*** (0.31)	-0.57*** (0.10)	0.36*** (0.05)	11,871	592
<i>Panel E. Host-country income tercile</i>						
Low	4.12*** (0.34)	4.64*** (0.37)	-0.45*** (0.07)	0.45*** (0.04)	17,581	1,004
Middle	4.07*** (0.23)	4.59*** (0.23)	-0.58*** (0.08)	0.41*** (0.03)	25,540	1,453
High	3.44*** (0.14)	3.85*** (0.15)	-0.46*** (0.06)	0.24*** (0.02)	48,479	2,268
<i>Panel F. Region</i>						
Europe	4.50*** (0.29)	5.08*** (0.29)	-0.68*** (0.08)	0.43*** (0.05)	17,532	815
Asia	3.55*** (0.25)	4.00*** (0.27)	-0.40*** (0.06)	0.38*** (0.04)	20,815	1,091
Oceania and South-West Pacific	3.83*** (0.35)	4.63*** (0.37)	-0.81*** (0.14)	0.39*** (0.09)	8,430	507
Africa	6.05*** (0.41)	6.67*** (0.46)	-0.75*** (0.12)	0.81*** (0.09)	9,141	536
Latin America	4.04*** (0.28)	4.70*** (0.32)	-0.63*** (0.16)	0.39*** (0.05)	13,272	739
Middle East	3.48*** (0.35)	4.08*** (0.40)	-0.59*** (0.15)	0.57*** (0.10)	2,911	152
North America	—	—	—	—	—	—
<i>Panel G. Topology-only regressors (log eigenvector, closeness, betweenness, degree)</i>						
Eigenvector centrality	0.376*** (0.012)	0.397*** (0.013)	-0.030*** (0.003)	0.011*** (0.001)	94,573	4,877
Closeness	4.40*** (0.16)	5.05*** (0.17)	-0.70*** (0.06)	0.51*** (0.03)	94,582	4,881
Betweenness	0.025*** (0.001)	0.025*** (0.001)	-0.001*** (0.000)	0.001*** (0.000)	94,508	4,881
Degree	0.605*** (0.016)	0.657*** (0.017)	-0.065*** (0.006)	0.046*** (0.003)	94,582	4,881
<i>Pooled interaction on log CO₂</i>						
Country's largest airport × log GACI	-1.93*** (0.32)				94,582	4,881
Global top 5 percent × log GACI	-2.57*** (0.20)				94,582	4,881

Each cell is the coefficient on log airport GACI (or on the topology component in Panel G) from a regression with airport and country-by-year fixed effects on the indicated subsample; standard errors clustered by airport. Country-by-year effects absorb every national shock, so the elasticities compare airports of the same country in the same year. The subsample of each country's largest airport is not separately estimable under country-by-year effects (one airport per country-year) and enters only through the interaction. Airport GACI embeds a capacity term; Panel G uses the topology-only components of the index. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Extended Data Table 7: Concentration of connectivity-attributed emissions across airports, 2023

	Top 1%	Top 5%	Top 10%	Top 25%	Gini
Emissions 2023	41.0	77.6	89.0	97.1	0.916
Attributed (country b)	45.9	84.6	94.2	99.3	0.945
Attributed (airport b)	46.8	85.3	94.6	99.4	0.947
Attributed (hub/non-hub b)	44.9	82.7	93.2	99.3	0.941

Shares (percent) of the global total held by the top-ranked airports, and Gini coefficients, over the 3,833 airports with positive 2023 emissions. Attributed emissions are CO₂ in 2023 times $1 - \exp(-\beta \Delta \ln \text{GACI}_{1996-2023})$ with β from the country 2SLS (5.35), the airport within-country regression (3.76), or the hub-specific airport estimates (2.11 for the global top 5 percent by 1996 capacity, 4.08 otherwise). Negative attributions (airports whose connectivity fell) are set to zero in the ranking.

Extended Data Table 8: Connectivity spillovers: own and neighbouring countries' connectivity, jointly instrumented

<i>Panel A. Log bunker CO₂; own and neighbour connectivity both instrumented</i>					
Neighbour definition	Own ln GACI	Neighbour ln GACI	SW <i>F</i> (own / nbr)	KP <i>F</i>	<i>N</i>
Contiguous countries (land border)	3.21*** (1.23)	1.85*** (0.66)	7 / 13	3.3	4,634
Five nearest countries	1.13 (1.78)	3.26* (1.82)	7 / 7	3.4	4,634
Countries within 500 km	4.29*** (1.22)	0.82 (0.62)	13 / 26	6.6	4,634
Kernel exp($-d/1000$ km)	-0.26 (1.94)	4.75** (2.25)	5 / 6	2.7	4,634
Inverse distance, all countries	-2.10 (4.32)	14.62 (9.22)	1 / 2	0.7	4,634
<i>Neighbour connectivity instrumented, own shifter in reduced form; s.e. clustered by country; AR = Anderson-Rubin 95% set</i>					
Neighbour definition	Own shifter (RF)	Neighbour ln GACI	AR set	KP <i>F</i>	<i>N</i>
Contiguous countries	controlled	1.96*** (0.61)	[0.8, 3.1]	168.0	4,634
Contiguous, sub-region $\times$ year FE	controlled	2.03** (0.83)	—	88.1	4,632
Countries within 500 km	controlled	0.34 (0.85)	[-1.3, 2.0]	89.0	4,634
Five nearest countries	controlled	6.69 (5.67)	[-4.0, 12.0]	1.4	4,634
Contiguous, joint 2SLS (own and neighbour instrumented)	own 3.21*** (1.23)	1.85*** (0.66)	[0.2, 3.0]	3.3 (SW 6.7 / 13.0)	4,634
Within 500 km, joint 2SLS	own 4.25*** (1.21)	0.86 (0.61)	[-0.4, 2.0]	6.7 (SW 13.5 / 32.1)	4,634
Inverse distance, own shifter in reduced form (previous specification) s.e. clustered by country	RF 0.25 (0.36)	6.73 (5.64) (5.64)	—	52.9	4,634
<i>Panel B. Margins of the response, contiguity definition, joint 2SLS</i>					
Outcome	Own ln GACI	Neighbour ln GACI		KP <i>F</i>	<i>N</i>
Bunker CO ₂	3.21*** (1.23)	1.85*** (0.66)		3.3	4,634
International CO ₂	2.82** (1.16)	2.06*** (0.59)		3.3	4,634
Domestic CO ₂	-2.78 (4.68)	1.96 (2.37)		2.7	3,567
Seat-km	2.82** (1.33)	2.45*** (0.66)		3.3	4,634
Flights	5.43*** (2.06)	1.16 (1.38)		3.3	4,634
Seats per flight	-0.69 (0.84)	0.30 (0.61)		3.3	4,634
Km per flight	-1.92 (1.23)	0.99 (0.73)		3.3	4,634
CO ₂ per seat-km	0.40 (0.40)	-0.60** (0.24)		3.3	4,634
International share of seat-km	-0.02 (0.16)	0.04 (0.09)		3.3	4,634

Notes: Neighbour connectivity is the leave-out average of other countries' log GACI under the stated weighting, instrumented by the identically weighted average of their Feyrer shifters; own connectivity is instrumented by the own shifter. All specifications include country and year fixed effects, log population, log sea market access and an indicator for countries with no neighbour under the definition. SW *F* is the Sanderson-Windmeijer conditional first-stage statistic for each endogenous regressor. The middle block of Panel A instruments neighbour connectivity only, with the own shifter as a control, and reports Anderson-Rubin 95 percent sets by grid inversion (subset sets in the joint rows; Methods). The last row uses all-country inverse-distance weights. Standard errors clustered by country in parentheses. ***, **, *: significance at 1, 5, and 10 percent.

Extended Data Table 9: Spatial structure of the connectivity spillover

<i>Panel A. One neighbour definition at a time (neighbour connectivity instrumented; own shifter in reduced form)</i>				
	Neighbour ln GACI	KP <i>F</i>	RMSE	<i>N</i>
Contiguous countries	1.96*** (0.61)	168.0		4,634
0–500 km	0.34 (0.85)	89.0		4,634
500–1,000 km	-0.30 (1.07)	40.6		4,634
1,000–2,000 km	2.45 (1.78)	13.4		4,634
2,000–5,000 km	2.27 (5.96)	5.0		4,634
Beyond 5,000 km	-4.34 (5.66)	621.3		4,634
Kernel exp($-d/\lambda$), $\lambda = 250$ km	2.07 (1.98)	6.7	0.4112	4,634
Kernel exp($-d/\lambda$), $\lambda = 500$ km	2.82 (2.16)	9.2	0.4083	4,634
Kernel exp($-d/\lambda$), $\lambda = 1,000$ km	4.09 (2.95)	15.0	0.4002	4,634
Kernel exp($-d/\lambda$), $\lambda = 2,000$ km	4.40 (5.07)	16.1	0.3942	4,634
Kernel exp($-d/\lambda$), $\lambda = 5,000$ km	5.05 (9.79)	38.7	0.3951	4,634
<i>Panel B. Regional exposure and region-by-year fixed effects</i>				
	Coefficient	KP <i>F</i>		<i>N</i>
Within aviation bloc (leave-out mean)	2.38*** (0.79)	36.0		4,634
Outside bloc (inverse-distance weighted), same regression	13.06** (5.35)			
Within UN sub-region (leave-out mean)	4.26 (4.52)	0.6		4,634
Outside sub-region, same regression	6.40 (15.66)			
Contiguous, joint 2SLS, sub-region $\times$ year FE	1.91*** (0.45) [1.32] {1.04}	3.4		4,632
Contiguous, own shifter in reduced form, sub-region $\times$ year FE	2.03** (0.83)	88.1		4,632
Kernel 500 km, own shifter in reduced form, sub-region $\times$ year FE	3.29 (3.03)	10.6		4,632
Inverse distance, own shifter in reduced form, sub-region $\times$ year FE	15.94* (9.17)	29.8		4,632
<i>Panel C. Permutation placebo: countries relabelled at random in the weight matrix (500 draws)</i>				
	Actual <i>t</i>	Permutation <i>p</i>	s.d. of permuted <i>t</i>	
Inverse distance (reduced form on the neighbour shifter)	3.46	0.188	2.64	
Contiguous countries (joint 2SLS, neighbour coefficient)	7.18	0.000	2.59	
Five nearest countries (joint 2SLS, neighbour coefficient)	5.35	0.018	2.36	

Notes: Panel A enters one neighbour exposure at a time, each instrumented by its own weighted shifter, with the own Feyrer shifter as a control; distance bands are leave-out means over countries whose aviation centroids fall in the stated range, with an indicator for empty bands. Panel B: aviation blocs are the EU/EEA, ASEAN, GCC, Mercosur, North America, ECOWAS, EAC/SADC, CIS, the Andean and Central American group, South Asia and Oceania; sub-regions follow the UN M49 classification. Panel C is the permutation placebo of the weight matrix (Methods); the *p*-value is the share of draws whose $|t|$ exceeds the actual. Standard errors clustered by country in parentheses (sub-region clusters in braces where shown). ***, **, *: significance at 1, 5, and 10 percent.

Extended Data Table 10: Spatial models under alternative weights and outcomes

	2SLS	SLX-IV	SAR-IV	SEM-IV	SDM-IV	SDM-IV
<i>Panel A. Inverse-distance W (all countries, 100 km floor); log bunker CO_2</i>						
ln GACI (own)	5.35*** (1.07)	-2.12 (4.34)	2.56*** (0.86)	4.01*** (0.75)	1.57 (1.38)	-1.93 (4.13)
W ln GACI (neighbours)	—	14.69 (9.27)	—	—	5.56 (7.51)	14.08 (8.59)
W ln CO_2 (spatial lag, ρ)	—	—	0.83*** (0.28)	—	0.16 (0.91)	—
W u (spatial error, λ)	—	—	—	0.35	—	-0.69
Direct effect	—	—	2.64*** (1.00)	—	1.59 (1.33)	—
Indirect effect	—	—	12.02 (42.77)	—	6.89 (80.90)	—
Total effect	—	—	14.67 (43.01)	—	8.48 (80.99)	—
First-stage F , own connectivity	18.5	1.5	3.5	9.2	2.2	0.7
<i>Panel B. Contiguity W; log international CO_2</i>						
ln GACI (own)	5.19*** (1.04)	2.83** (1.15)	5.26*** (0.90)	4.75*** (1.09)	4.59*** (0.93)	2.77*** (0.99)
W ln GACI (neighbours)	—	2.05*** (0.59)	—	—	1.18** (0.60)	2.12*** (0.60)
W ln CO_2 (spatial lag, ρ)	—	—	0.03 (0.04)	—	-0.02 (0.01)	—
W u (spatial error, λ)	—	—	—	0.07	—	0.07
Direct effect	—	—	5.26*** (0.98)	—	4.58*** (0.95)	—
Indirect effect	—	—	0.14 (0.17)	—	0.86** (0.43)	—
Total effect	—	—	5.40*** (1.02)	—	5.44*** (0.86)	—
First-stage F , own connectivity	18.5	7.2	5.2	7.9	5.3	5.1
<i>Panel C. Contiguity W; log CO_2 per seat-km</i>						
ln GACI (own)	-0.29 (0.29)	0.39 (0.40)	-0.27 (0.26)	-0.18 (0.29)	-0.26 (0.26)	0.26 (0.37)
W ln GACI (neighbours)	—	-0.60** (0.23)	—	—	-0.01 (0.20)	-0.51** (0.22)
W ln CO_2 (spatial lag, ρ)	—	—	0.14* (0.08)	—	0.14 (0.11)	—
W u (spatial error, λ)	—	—	—	0.09	—	0.01
Direct effect	—	—	-0.27 (0.25)	—	-0.26 (0.23)	—
Indirect effect	—	—	-0.03 (0.04)	—	-0.05 (0.17)	—
Total effect	—	—	-0.30 (0.28)	—	-0.31 (0.28)	—
First-stage F , own connectivity	18.5	7.2	5.1	7.6	4.3	4.7
<i>Panel D. Inverse-distance W; log international CO_2</i>						
ln GACI (own)	5.19*** (1.04)	-0.75 (3.42)	2.70*** (0.86)	4.15*** (0.73)	1.84 (1.31)	-1.43 (3.66)
W ln GACI (neighbours)	—	11.69 (7.44)	—	—	5.56 (8.32)	12.91* (7.69)
W ln CO_2 (spatial lag, ρ)	—	—	0.81*** (0.30)	—	0.08 (1.11)	—
W u (spatial error, λ)	—	—	—	0.27	—	-0.42
Direct effect	—	—	2.78*** (0.86)	—	1.85 (1.52)	—
Indirect effect	—	—	11.36 (40.79)	—	6.17 (130.34)	—
Total effect	—	—	14.14 (40.82)	—	8.02 (130.49)	—
First-stage F , own connectivity	18.5	1.5	3.3	9.8	2.2	0.7
<i>Panel E. Inverse-distance W; log CO_2 per seat-km</i>						
ln GACI (own)	-0.29 (0.29)	1.06 (1.14)	-0.04 (0.24)	-0.01 (0.26)	-0.23 (0.37)	1.21 (1.17)
W ln GACI (neighbours)	—	-2.66 (2.53)	—	—	0.50 (0.84)	-2.91 (2.74)
W ln CO_2 (spatial lag, ρ)	—	41	0.94 (0.72)	—	1.15 (0.73)	—
W u (spatial error, λ)	—	—	—	0.56	—	0.19
Direct effect	—	—	-0.05	—	-0.24	—

Extended Data Table 11: Exclusion-restriction diagnostics for the Feyrer instrument

	Reduced form	2SLS	KP F	N
<i>Panel A. Outcomes: coefficient on the instrument (reduced form) and on log GACI (2SLS)</i>				
Aviation CO ₂ (total, our series)	0.685*** (0.149)	5.35*** (1.08)	18.4	4,634
Aviation CO ₂ (international)	0.665*** (0.148)	5.19*** (1.04)	18.4	4,634
Territorial fossil CO ₂ excl. domestic aviation (GCP/OWID)	0.049 (0.120)	0.39 (0.95)	17.1	4,576
Oil CO ₂ excl. domestic aviation (OWID)	-0.115 (0.110)	-0.91 (0.89)	17.1	4,576
Transport CO ₂ excl. domestic aviation (EDGAR)	0.111 (0.117)	0.90 (0.94)	15.8	4,461
Power industry CO ₂ (EDGAR)	-0.365 (0.288)	-2.96 (2.51)	15.3	4,409
Buildings CO ₂ (EDGAR)	-0.140 (0.153)	-1.13 (1.23)	15.8	4,460
Industrial combustion CO ₂ (EDGAR)	0.182 (0.168)	1.53 (1.48)	14.6	4,432
Coal CO ₂ (OWID)	0.896*** (0.314)	15.72 (10.06)	2.5	3,030
Gas CO ₂ (OWID)	-0.333 (0.316)	-4.28 (4.98)	3.7	2,863
Cement CO ₂ (OWID)	0.665*** (0.232)	6.07** (2.70)	8.9	3,476
Agriculture CO ₂ (EDGAR)	-0.421* (0.221)	-4.06 (2.74)	6.5	2,667
<i>Panel B. Content of the global cycle: first stage of the aviation-technology instrument with rival cycle $\times$ geography added</i>				
	First-stage π	2SLS	Partial F	N
Aviation technology $\times$ air geography (baseline)	0.128*** (0.030)	5.35*** (1.08)	18.4	4,634
plus oil price $\times$ geography	0.103*** (0.027)	4.84*** (1.15)	14.7	4,634
plus world GDP $\times$ geography	0.011 (0.028)	-2.64 (11.92)	0.2	4,634
plus world exports $\times$ geography	-0.005 (0.028)	28.44 (162.69)	0.0	4,634
plus all three	-0.013 (0.026)	13.30 (25.58)	0.2	4,634
World GDP $\times$ geography alone	0.170*** (0.037)	RF 0.950*** (0.197)	21.4	4,634
Oil price $\times$ geography alone	0.103*** (0.023)	RF 0.631*** (0.128)	20.0	4,634
World exports $\times$ geography alone	0.169*** (0.036)	RF 0.947*** (0.195)	21.9	4,634
<i>Panel C. Placebo instrument: aviation technology $\times$ 1996 sea market access</i>				
	First-stage π	Reduced form	Partial F	N
Sea-geography instrument alone	0.117*** (0.033)	0.482*** (0.171)	12.6	4,634
Sea-geography instrument, air-geography instrument included	0.033 (0.051)	-0.187 (0.257)	0.4	4,634
Air-geography instrument, sea-geography instrument included	0.104** (0.044)	0.824*** (0.213)	5.6	4,634
2SLS with the sea interaction as a control		7.95*** (2.90)	5.6	4,634
<i>Panel D. Zero-first-stage test by baseline connectivity tercile</i>				
	First-stage π	Reduced form (CO ₂)	Partial F	N
Low baseline connectivity	0.263*** (0.065)	1.966*** (0.351)	16.4	1,453
Middle	0.081* (0.042)	0.410* (0.228)	3.7	1,526
High (instrument does not move GACI)	0.024 (0.052)	0.095 (0.186)	0.2	1,655
<i>Panel E. Development-channel controls added sequentially (2SLS coefficient on log GACI)</i>				
	2SLS		KP F	N
Baseline (log population, log sea market access)	5.35*** (1.08)		18.4	4,634
plus log GDP	4.45*** (1.13)		11.8	4,634
plus trade/GDP	4.43*** (1.32)		9.6	4,618
plus urban share	4.35*** (1.30)		10.4	4,618
plus FDI/GDP	4.54*** (1.20)		12.0	4,432
plus log tourist arrivals	4.64*** (1.40)		9.8	3,460
Baseline on the common sample of the last row	5.62*** (1.25)		18.0	3,460

All regressions include country and year fixed effects, log population and log sea market access; standard errors clustered by country in parentheses. Panel A: reduced form of each log outcome on the Feyrer interaction; non-aviation series are Global Carbon Project territorial CO₂ (via Our World in Data; domestic aviation subtracted) and EDGAR v2024 sectoral CO₂. Panel B: rival cycles are world real GDP, the Brent price and world real exports, min-max scaled and interacted with log 1996 air market access (correlations with the aviation index 0.82, 0.53 and 0.84). Panel C replaces air by sea market access in the interaction. Panel D: terciles of 1996 capacity-weighted GACI. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Extended Data Table 12: Functional form of the connectivity–emissions relationship

	2SLS	OLS	Implied elasticity at sample means	KP <i>F</i>
<i>Panel A. Full sample</i>				
Log CO ₂ on log GACI (reference)	5.347*** (1.076)	3.388*** (0.364)	5.35 (elasticity)	18.4
Log CO ₂ on GACI level (semi-log)	3.859*** (0.952)	2.055*** (0.292)	4.19 ($\beta \times$ mean GACI (1.09))	18.9
CO ₂ (Mt) on GACI level (level-level)	4.295 (3.999)	11.092*** (3.655)	1.22 ($\beta \times$ mean GACI / mean CO ₂ (3.81 Mt))	18.9
CO ₂ (Mt) on log GACI (lin-log)	5.951 (5.798)	–	1.56 (β / mean CO ₂)	18.4
<i>Panel B. Semi-log and level specifications by baseline-connectivity tercile</i>				
	Semi-log β	Level β (Mt per unit)	Implied elasticities (semi-log; level)	KP <i>F</i>
Low baseline connectivity	9.077*** (2.234)	1.294*** (0.311)	6.93; 6.78 (mean GACI 0.76, mean CO ₂ 0.15 Mt)	14.3
Middle	4.591* (2.593)	8.616* (4.498)	4.46; 11.22 (mean GACI 0.97, mean CO ₂ 0.75 Mt)	3.2
High	1.103 (1.738)	-27.546 (38.140)	1.63; -4.11 (mean GACI 1.47, mean CO ₂ 9.87 Mt)	1.3

Notes: Specification as in Table 1, column 1, with the connectivity regressor and the outcome transformed as stated; the instrument is the Feyrer interaction throughout. GACI (capacity-weighted mean) has a sample mean of 1.09 and ranges from 0.56 to 2.57 at baseline; a one-unit change is therefore roughly a doubling for a typical country. Implied elasticities evaluate the semi-log and level slopes at sample means. Standard errors clustered by country in parentheses. ***, **, *: significance at 1, 5, and 10 percent.

Extended Data Table 13: The elasticity along the baseline-connectivity distribution

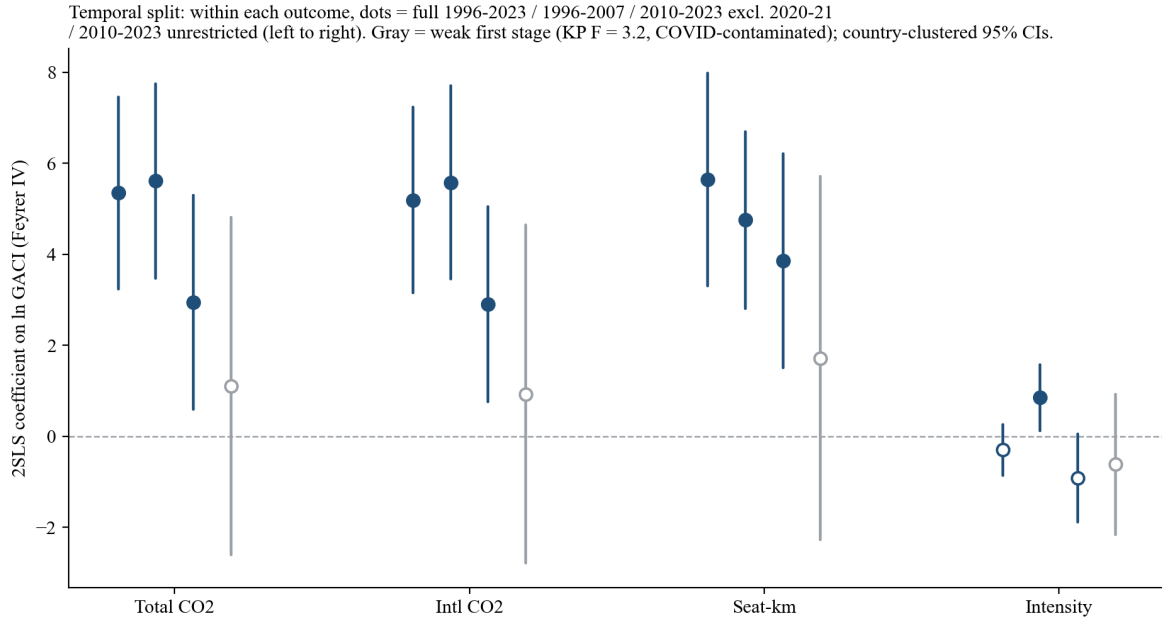
Panel A. Split-sample 2SLS by quintile of 1996 connectivity						
Quintile	Baseline GACI (range)	Log-log elasticity	KP <i>F</i>	Semi-log slope	Implied elasticity	<i>N</i>
1	0.68 (0.56–0.74)	7.37*** (1.72)	13.3	9.14*** (2.33)	6.65	900
2	0.79 (0.74–0.84)	8.50*** (2.94)	10.9	8.96*** (2.91)	7.53	857
3	0.91 (0.84–1.00)	3.30 (2.31)	4.8	2.71 (2.07)	2.64	951
4	1.08 (1.00–1.17)	84.74 (8451.67)	0.0	3.61 (12.62)	4.24	958
5	1.56 (1.18–2.57)	4.01 (2.54)	2.1	1.41* (0.85)	2.33	968
Panel B. Pooled 2SLS with log GACI interacted with centred log baseline GACI (instruments interacted identically)						
	Linear interaction		Quadratic interaction			
ln GACI	5.52*** (1.04)		5.12*** (0.94)			
ln GACI × (ln baseline GACI – mean)	-3.23*** (1.05)		-3.84** (1.51)			
ln GACI × (ln baseline GACI – mean) ²	–		2.63 (2.62)			
KP <i>F</i> ; <i>N</i>	9.2; 4,634		6.8; 4,634			
Panel C. Implied elasticity at percentiles of baseline connectivity (delta-method s.e.)						
	p10	p25	p50	p75	p90	
Linear	6.52 (1.17)	6.17 (1.12)	5.60 (1.05)	5.00 (1.01)	4.09 (1.01)	
Quadratic	6.57 (1.20)	6.00 (1.04)	5.22 (0.94)	4.56 (0.93)	3.93 (0.97)	
Baseline GACI at percentile	0.70	0.78	0.93	1.12	1.48	

Notes: Baseline connectivity is the earliest observed capacity-weighted GACI of each country. Panel A estimates Table 1, column 1, within quintiles; the implied elasticity of the semi-log specification is the slope times the quintile's mean GACI. Panel B interacts log GACI with the centred log baseline value (mean -0.05), instrumenting each term with the Feyrer interaction times the same centred value; Panel C evaluates the fitted elasticity at percentiles of the baseline distribution. Standard errors clustered by country in parentheses. ***, **, *: significance at 1, 5, and 10 percent.

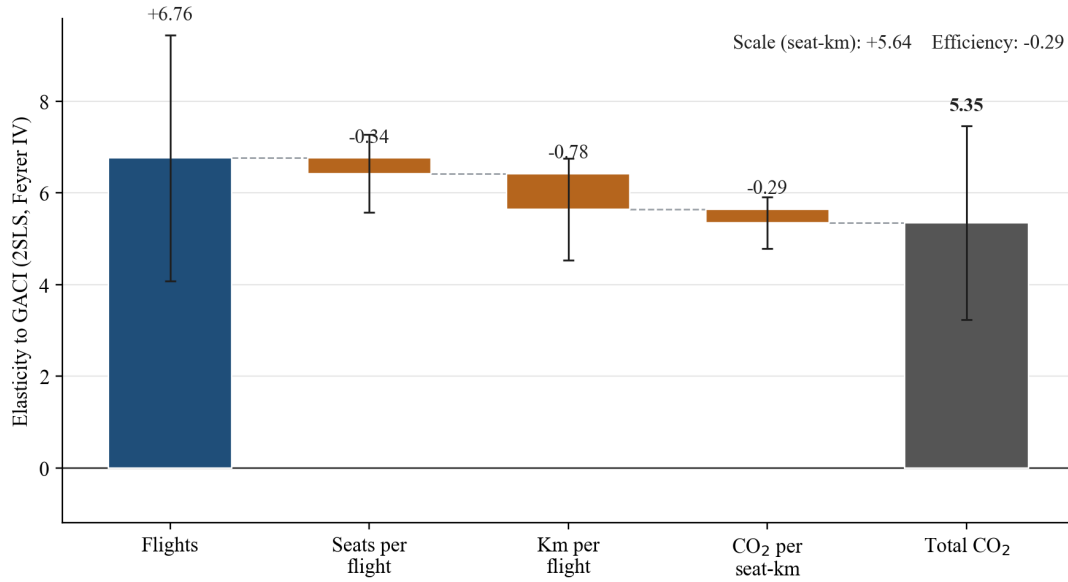
Extended Data Table 14: Sensitivity of the 2023 attribution to heterogeneous elasticities

Elasticity rule	Attributed 2023 CO ₂ (Mt)	Share of 2023 emissions (%)
Common elasticity (5.35)	348.7	41.1
Baseline-connectivity terciles (8.46 / 5.07 / 4.05)	311.4	36.7
Baseline-income terciles (10.40 / 6.14 / 1.83)	294.2	34.7
Continuous, linear interaction	321.2	37.9
Continuous, quadratic interaction	315.7	37.2
Semi-log slope (3.86 per GACI unit) on the absolute change	346.0	40.8
<i>Largest contributors under the common and the continuous (quadratic) rule, Mt</i>		
CHN (baseline GACI 1.16)	85.9	79.9
USA (baseline GACI 2.01)	34.5	24.6
ARE (baseline GACI 1.58)	24.5	22.8
JPN (baseline GACI 1.32)	16.9	14.4
TUR (baseline GACI 1.21)	16.5	15.8
IND (baseline GACI 1.18)	15.4	13.7
KOR (baseline GACI 0.88)	14.1	14.1
CAN (baseline GACI 1.48)	12.7	10.6

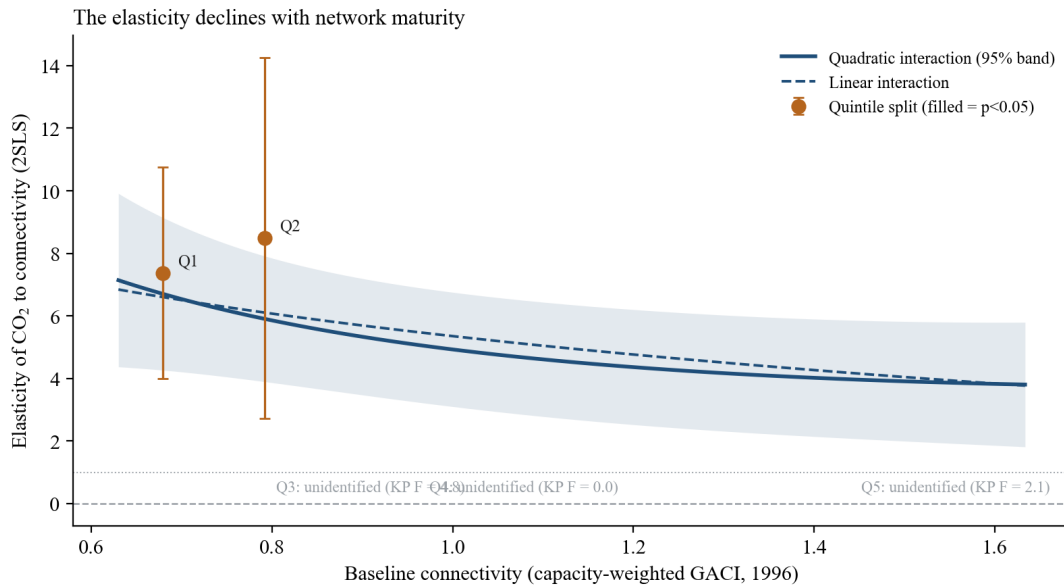
Notes: Attributed share = $1 - \exp(-\beta_c \Delta \ln \text{GACI}_c)$ applied to 2023 bunker emissions, with β_c assigned by the stated rule; terciles use the split-sample estimates of Extended Data Table 3 (the top connectivity tercile is weakly identified), and the continuous rules use the fitted elasticity of Extended Data Table 13 at each country's baseline, truncated at zero. The semi-log rule applies the slope to the absolute change in GACI. Shares are of world 2023 aviation CO₂ (848 Mt; the 184-country sample holds 833 Mt).



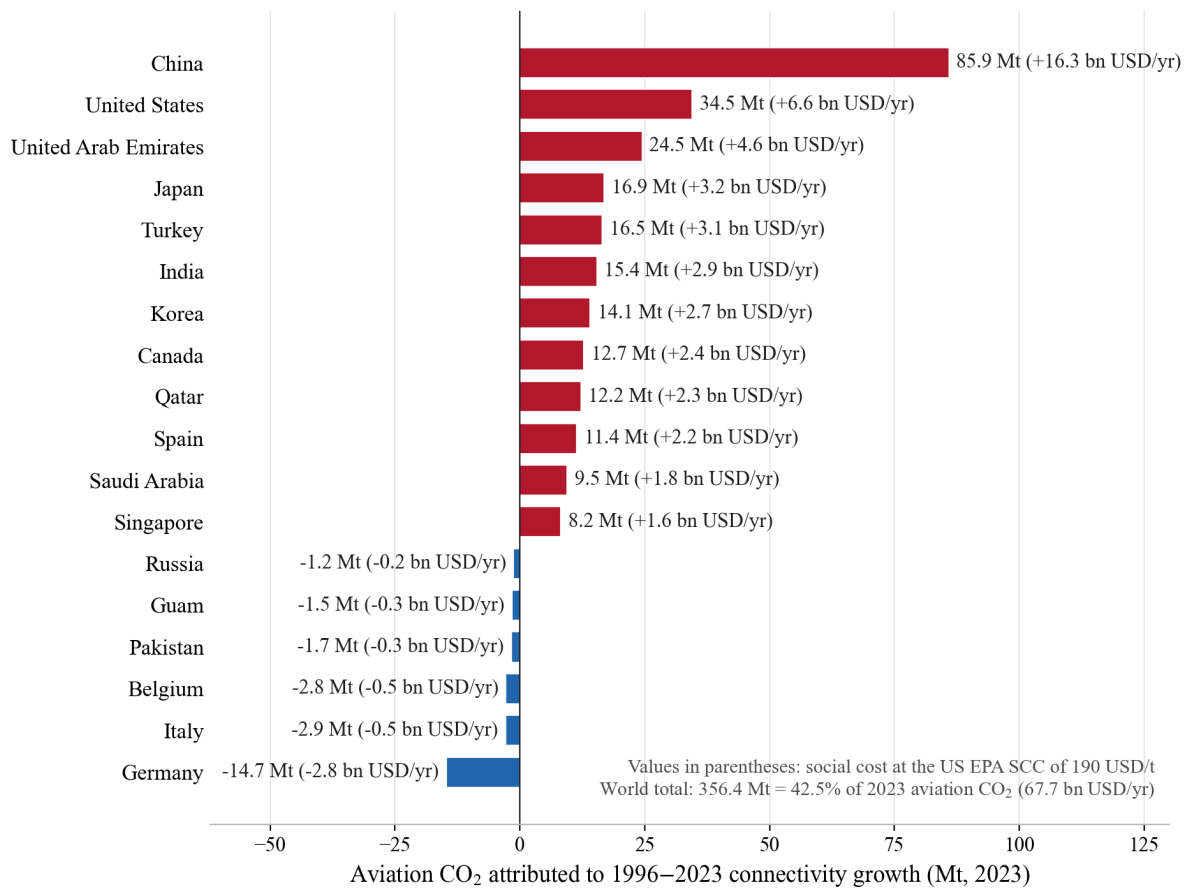
Extended Data Fig. 1: Temporal split of the connectivity elasticity. Within each outcome, markers show the full sample, the network-expansion era (1996–2007), the mature-network era (2010–2023 excluding the COVID-19 years 2020–2021), and the unrestricted 2010–2023 sample from left to right; gray denotes the unrestricted later sample, whose first stage (KP $F = 3.2$) is contaminated by the COVID collapse. Bars are 95 percent confidence intervals with standard errors clustered by country.



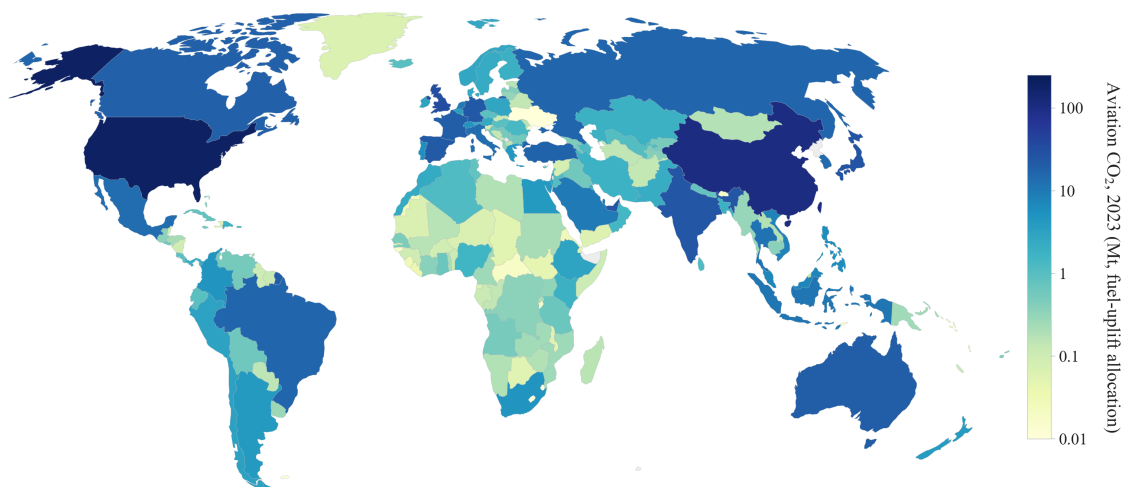
Extended Data Fig. 2: **Decomposition of the CO₂ elasticity on the accounting identity.** Each bar is the 2SLS elasticity of the component with respect to connectivity (Feyrer instrument; 95 percent confidence intervals); the components add exactly to the total. Scale is the sum of the first three bars.



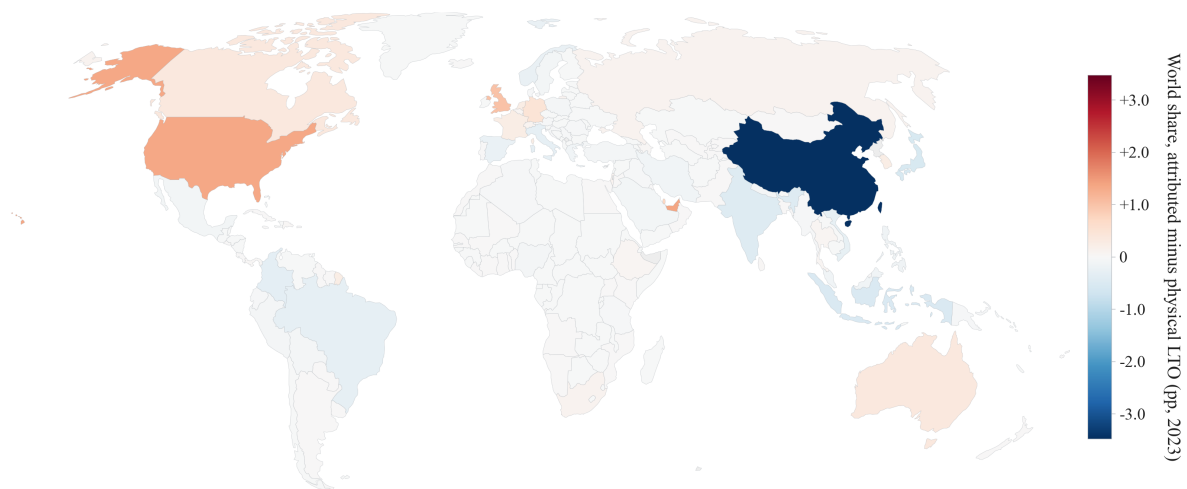
Extended Data Fig. 3: **The elasticity along the baseline-connectivity distribution.** Quintile split-sample 2SLS estimates (points, 95 percent confidence intervals; the fourth quintile is unidentified and omitted) and the fitted elasticity from pooled specifications that interact log GACI with centred log baseline GACI (linear, dashed; quadratic with 95 percent band). Horizontal lines mark zero and one.



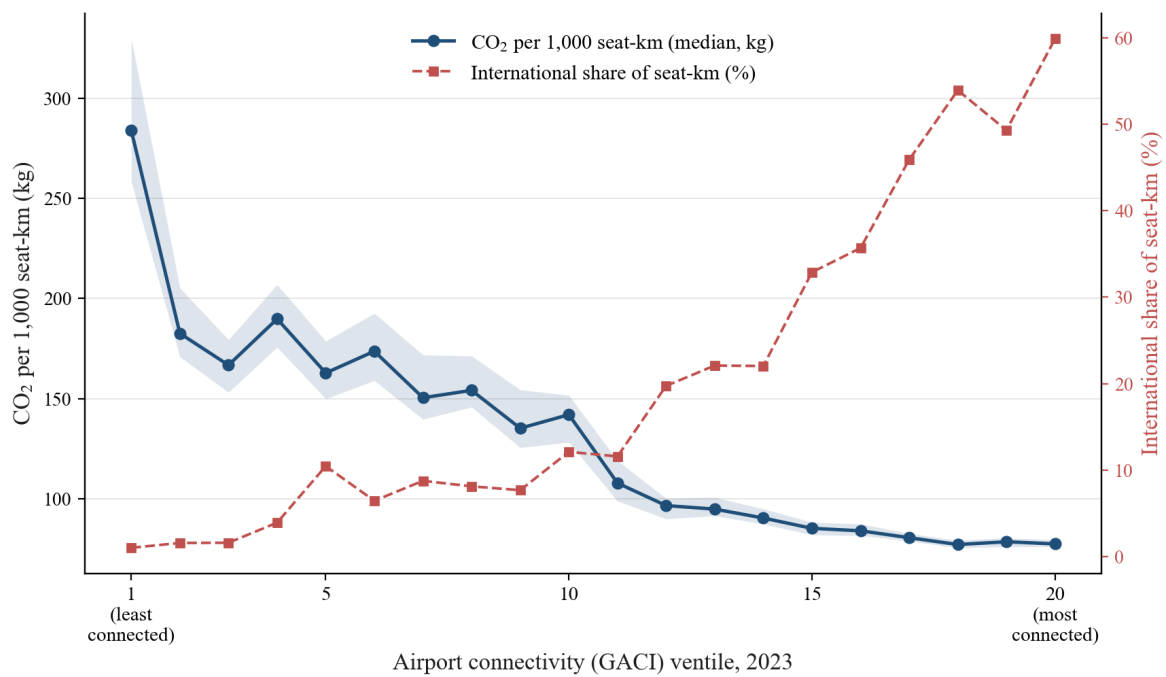
Extended Data Fig. 4: **Country-level aviation CO₂ attributed to 1996–2023 connectivity growth, with its social cost.** Top twelve positive countries and all countries below –1 Mt; parenthetical values apply the US EPA social cost of carbon (\$190 per tonne). The world total is 348.7 Mt, 41.1 percent of 2023 aviation CO₂ (\$66.3 billion per year).



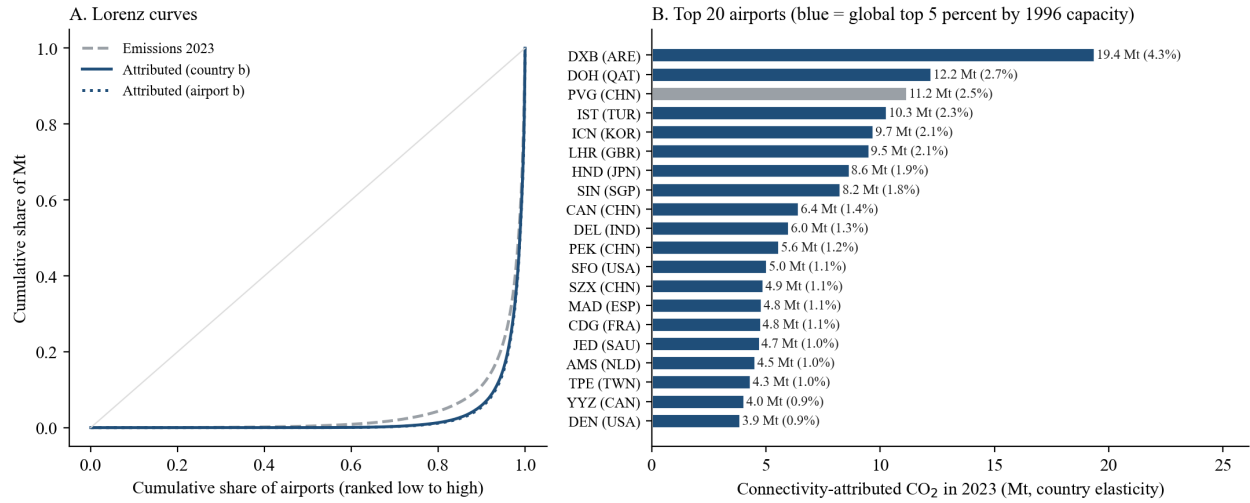
Extended Data Fig. 5: **Aviation CO₂ levels under the bunker convention, 2023.**



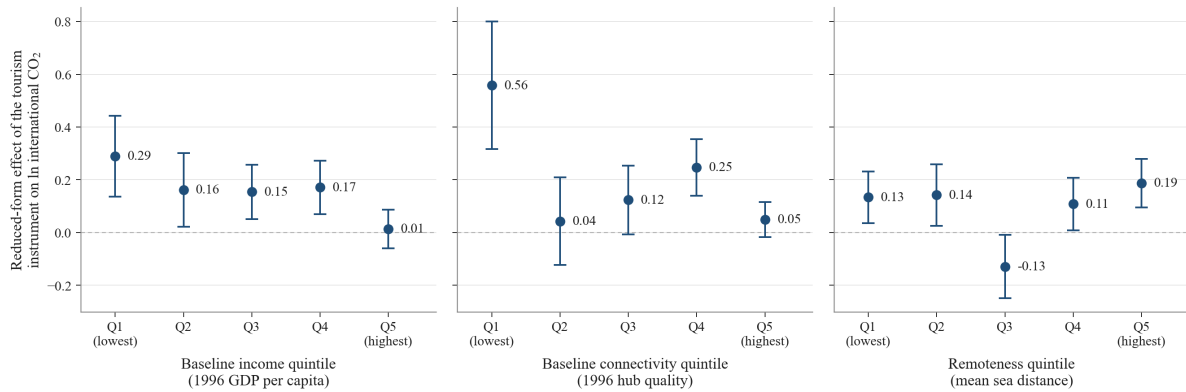
Extended Data Fig. 6: **Attribution mismatch, 2023:** bunker-attributed share minus physical LTO share (percentage points). Positive values indicate long-haul, cruise-heavy networks; negative values indicate short-haul, domestically oriented networks.



Extended Data Fig. 7: **Airport-level efficiency curve, 2023:** median kg CO₂ per 1,000 seat-km by connectivity ventile, with the international share of seat-km on the second panel.

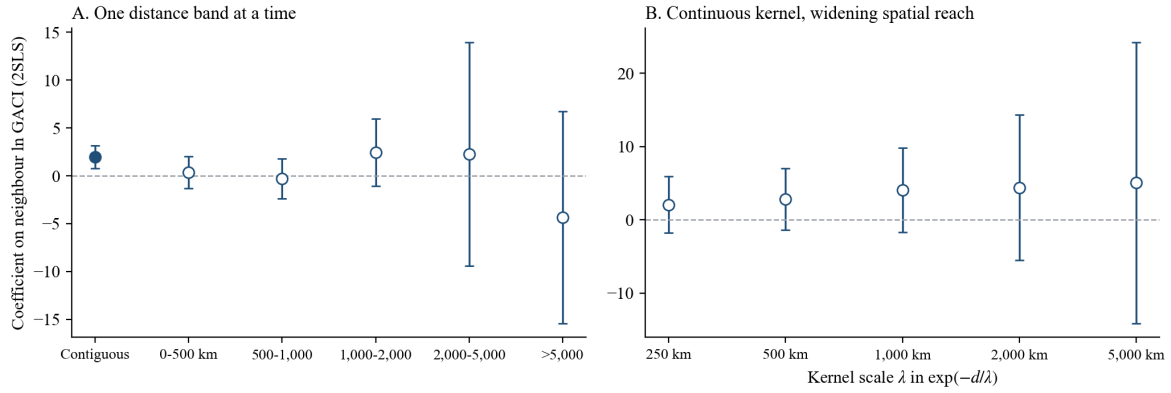


Extended Data Fig. 8: **Concentration of connectivity-attributed emissions across airports, 2023.** A, Lorenz curves of 2023 emissions and of emissions attributed to 1996–2023 connectivity growth over 3,833 airports with positive emissions. B, the twenty largest attributed contributions (Mt and share of the world attributed total); blue marks airports in the global top five percent by 1996 capacity.

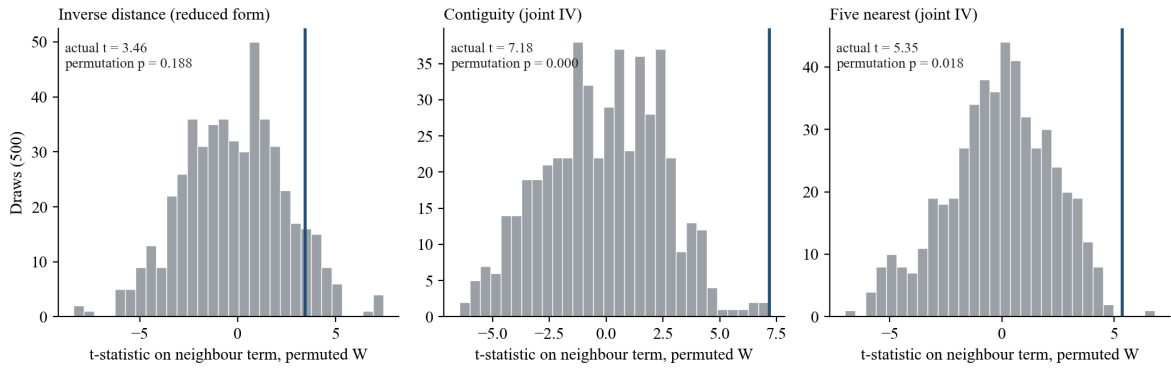


Extended Data Fig. 9: Reduced-form quintile slopes under the tourism shifter (heteroskedasticity-robust 95 percent confidence intervals): effects are concentrated in low-income, low-connectivity quintiles.

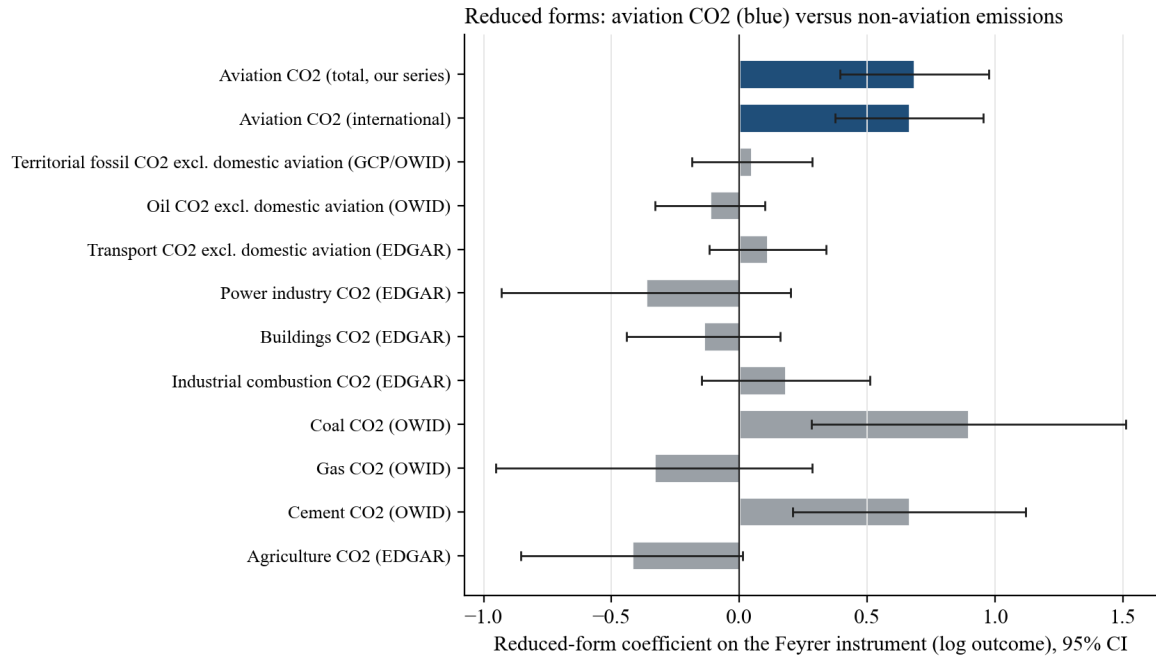
Spatial profile of the spillover (filled = $p < 0.05$; bars = 95% CI; own shifter controlled)



Extended Data Fig. 10: Spatial profile of the spillover. A, coefficient on neighbour connectivity when one distance band at a time defines the neighbours (contiguous countries, then bands of aviation-centroid distance); B, the same with an exponential kernel of widening scale. Neighbour connectivity is instrumented by the identically weighted shifter and the own shifter is controlled. The absence of a smooth decline and the negative but imprecise coefficient beyond 5,000 km indicate that wide exposures track global trends rather than a spatial spillover.



Extended Data Fig. 11: Permutation placebo for the spillover. Distributions of the t -statistic on the neighbour term over 500 random relabellings of countries in the weight matrix, with the actual statistic in blue, for the inverse-distance reduced form and for the contiguity and five-nearest joint 2SLS specifications.



Extended Data Fig. 12: Reduced forms of the Feyrer instrument on aviation CO₂ (blue) and on non-aviation emission series (grey), with 95 percent confidence intervals. Specification as in Table 1.

Supplementary Tables

Supplementary Table 1: Alternative instrument constructions, leads, overidentification and inference

	First-stage π	2SLS	KP F	N
<i>Panel A. Technology series in the interaction (all $\times$ log 1996 air market access)</i>				
World seat capacity (baseline)	0.128*** (0.030)	5.35*** (1.08)	18.4	4,634
World flights	0.128*** (0.031)	5.25*** (1.17)	17.2	4,634
World seat-km	0.147*** (0.033)	5.38*** (1.05)	19.6	4,634
World sum of GACI	0.137*** (0.031)	5.64*** (1.02)	19.5	4,634
World fuel-efficiency index (CO ₂ per seat-km, inverted)	0.160*** (0.035)	5.62*** (0.98)	21.3	4,634
Baseline geography recomputed ($\theta = 1$)	0.127*** (0.029)	5.55*** (1.10)	19.3	4,634
Distance decay $\theta = 0.5$	0.274*** (0.064)	5.36*** (1.09)	18.2	4,634
Distance decay $\theta = 1.5$	0.073*** (0.017)	5.82*** (1.18)	18.0	4,634
<i>Panel B. Leads of the instrument (reduced form on log CO₂)</i>				
	Current	Lead		N
Three-year lead	0.548*** (0.118)	0.390*** (0.096)		4,078
Five-year lead	0.684*** (0.164)	0.301*** (0.076)		3,715
<i>Panel C. Overidentification and inference (2SLS on log CO₂)</i>				
	2SLS	s.e.	KP F	N
Feyrer and tourism-heritage instruments jointly	4.72***	(0.88)	18.7	4,634
Hansen J (p-value)	3.46 (0.063)			
Tourism-heritage instrument alone	1.39	(1.59)	8.3	4,634
Heteroskedasticity-robust	5.35***	(0.40)	153.6	4,634
Clustered by country	5.35***	(1.08)	18.4	4,634
Two-way clustered (country, year)	5.35***	(1.07)	15.6	4,634

Specification as in Table 1, column 1. Panel A replaces the world seat-capacity index by other min-max scaled world aviation series, or recomputes 1996 air market access with a different distance-decay exponent. Panel B enters the instrument dated $t + 3$ or $t + 5$ alongside its current value; because the technology index is a smooth global trend, the lead is mechanically collinear with the current value and this test is informative only about differential trends by geography. Panel C: the tourism-heritage instrument fails the size-by-cycle stress test (Methods) and is shown for reference; the Hansen test rejects equality of the two just-identified estimates. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Supplementary Table 2: Airport-level identification pilot with an airport Feyrer shifter

	Coefficient	Partial / KP F	N
<i>First stage: log airport GACI on $a_t \times$ log airport air market access 1996</i>			
Baseline	-0.018 (0.027)	0.5	77,591
plus $a_t \times$ log 1996 seat-km	0.021 (0.026)	0.7	77,539
plus $a_t \times$ log 1996 GACI	0.015 (0.025)	0.3	77,539
<i>2SLS (baseline instrument)</i>			
log CO ₂	1.52 (10.95)	0.5	77,591
log seat-km	8.68 (14.49)	0.4	77,501
log CO ₂ /seat-km	1.91 (6.09)	0.4	77,501
International share	2.06 (3.36)	0.4	77,501

Airport air market access is the population-weighted inverse distance from the airport to all foreign countries' aviation centroids in 1996 (own country excluded); a_t is the world seat-capacity index of the country instrument. Airport and country-by-year fixed effects, standard errors clustered by airport. Within a country the shifter varies only through airports' relative location (mean within-country standard deviation of the log 1996 access term 0.04), and it has no first stage; the pilot is therefore not used and airport-level estimates are reported as within-country descriptive evidence.

Supplementary Table 3: Aviation-accident instrument (Aviation Safety Network), pilot

	Bunker CO ₂	Intl. CO ₂	Intensity	KP F
2SLS, Lagged fatal accidents	5.59** (2.25)	4.57** (1.80)	-0.92 (0.65)	4.4
2SLS, Lagged accident deaths	5.26* (3.12)	4.24 (2.80)	-0.28 (0.99)	2.0
2SLS, Lagged any-fatal-accident indicator	6.05 (4.31)	5.34 (3.98)	-1.02 (1.49)	1.2
Feyrer and lagged fatal accidents jointly	5.09*** (0.40)	4.90*** (0.39)	-0.34*** (0.12)	66.8
Hansen J p -value	0.81	0.86	0.27	
Feyrer and lagged accident deaths jointly	5.08*** (0.40)	4.89*** (0.39)	-0.33*** (0.12)	66.3
Hansen J p -value	0.95	0.82	0.96	

Notes: Country-year accident counts and deaths from the Aviation Safety Network, lagged one year, as instruments for log GACI; specification otherwise as in Table 1. The accident instruments are weak (first-stage F of 1 to 4) and are not used on their own; the joint specification with the Feyrer instrument is reported for the overidentification test. Standard errors clustered by country in parentheses. ***, **, *: significance at 1, 5, and 10 percent.

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